

인플레이션 헤지 전략

US CPI Urban Consumers Core MoM SA

□ CPI



From Bloomberg

Core CPI does not include the costs of goods and services in the food and energy sectors, the prices of which are very volatile.

Seasonal adjustments are made to eliminate the effect of regular seasonal movements from climatic conditions, vacation practices, and similar factors.

인플레이션, 주식과 채권 수익률

- 2,558개의 국가-연도 동안 높은 인플레이션과 낮은 인플레이션 기간 동안의 실질 주식 및 채권 수익률

Figure 16: Real bond and equity returns versus inflation rates, 1900–2021



Source: Elroy Dimson, Paul Marsh, and Mike Staunton, DMS Database 2022, Morningstar. Not to be reproduced without express written permission from the authors.

왜 사람들은 경기 침체가 끝날 무렵 인플레이션을 걱정할까?

- 왜 사람들은 경기 침체가 끝날 무렵 인플레이션을 걱정할까?
- 막대한 정부 적자 때문이다.

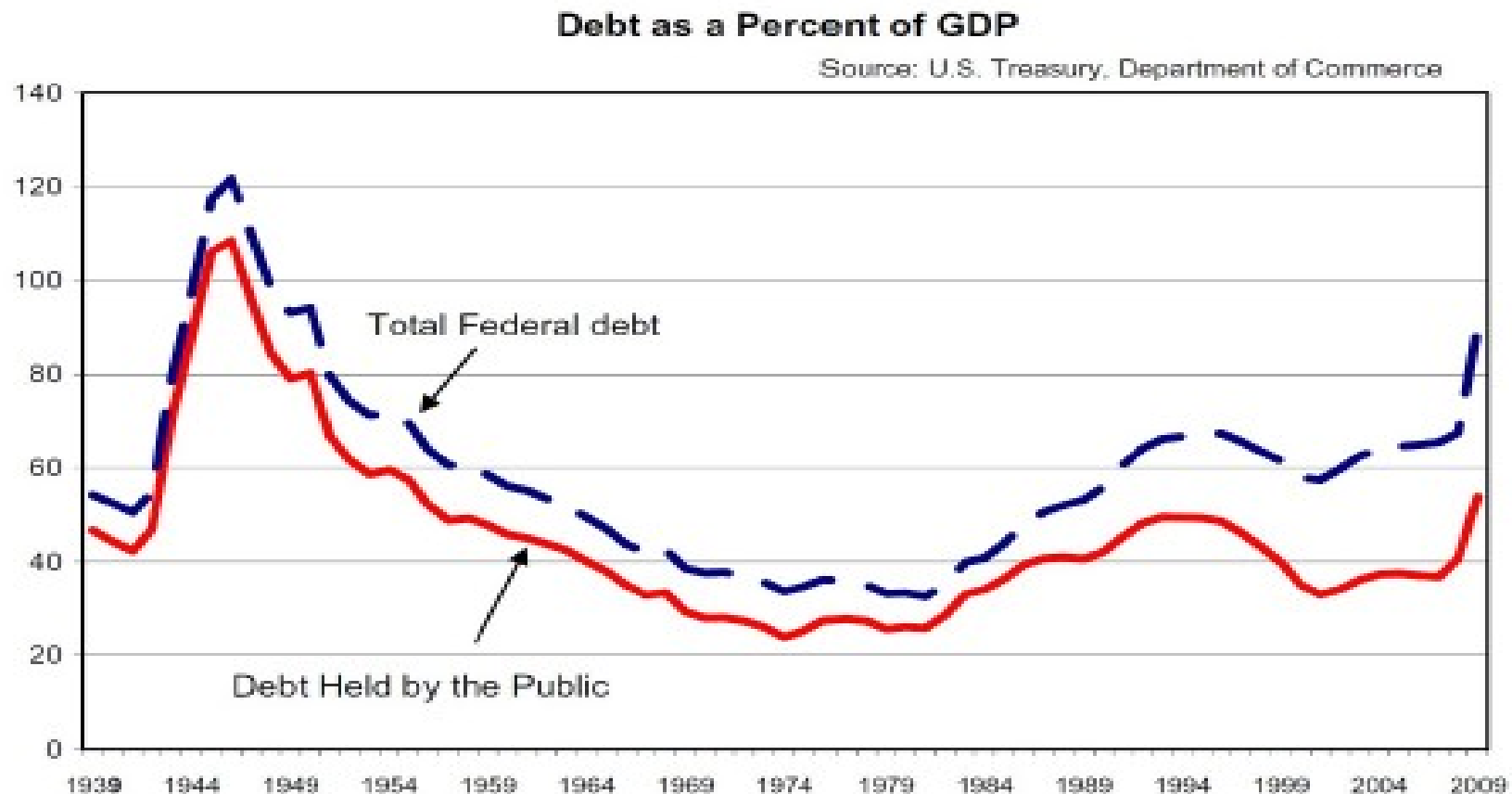


Fig. 1 Debt as a share of GDP

Aizenman and Marion, Journal of Macroeconomics, 2011

재정 정책: 2009년 전망

□ 2009년 전망

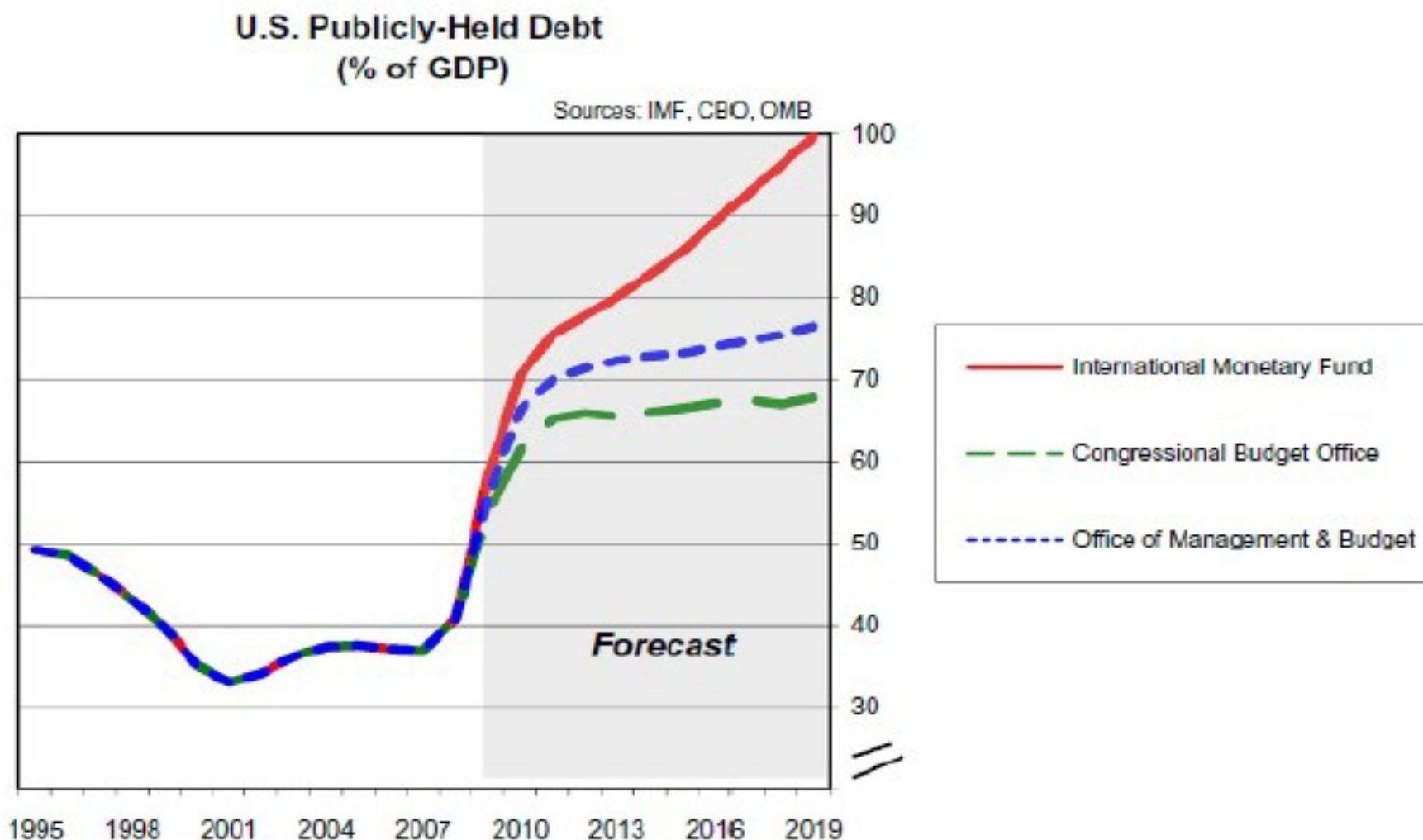


Fig. 3. America's projected debt burden.

막대한 정부 부채를 어떻게 처리해야 할까?

- 막대한 정부 부채를 어떻게 처리할까?
- 하나의 방법은 인플레이션을 통해 해결하는 것이다.

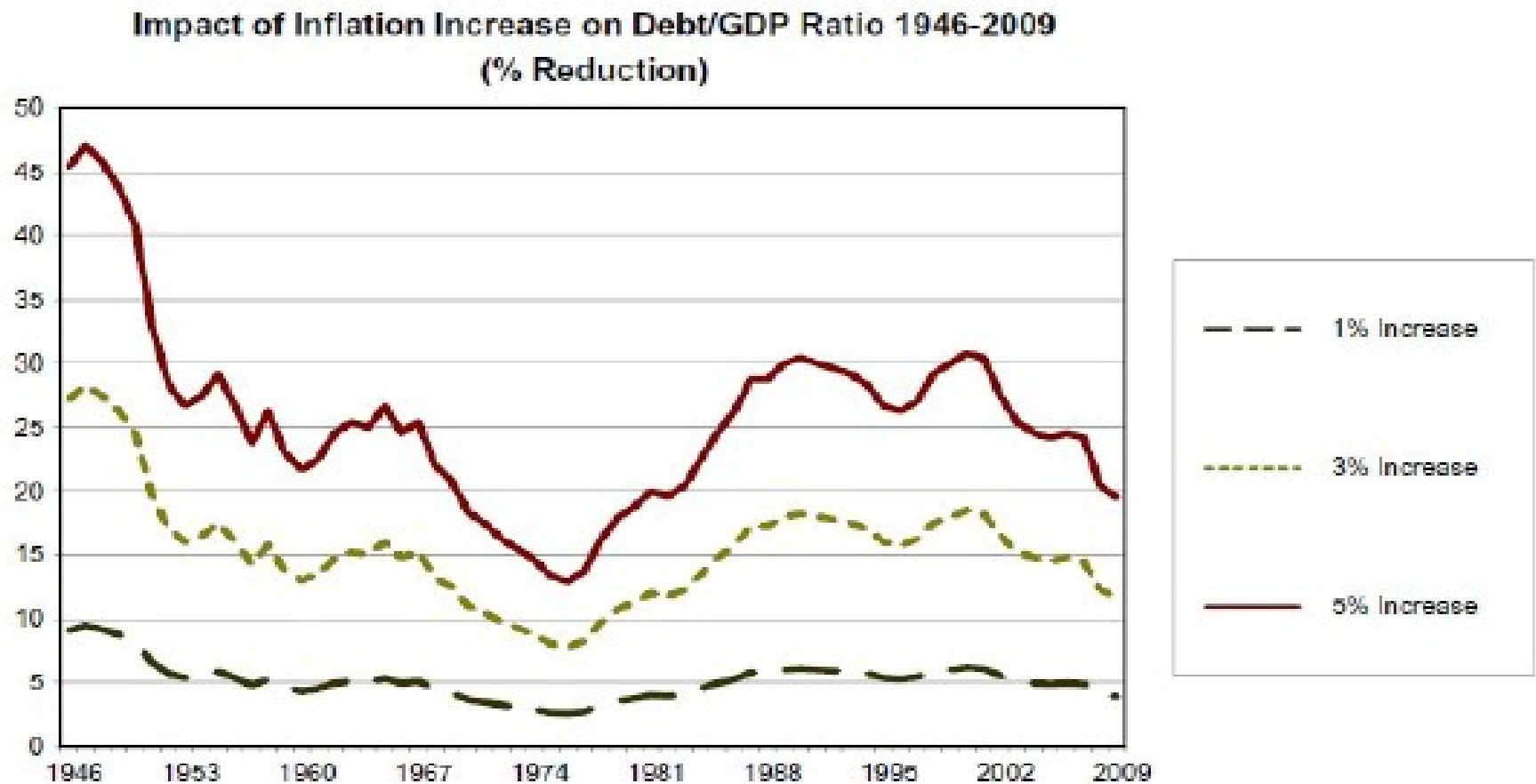


Fig. 9. Impact of inflation on publicly-held debt as a share of GDP.

Aizenman and Marion, Journal of Macroeconomics, 2011

연준은 인플레이션을 어떻게 통제하려고 하는가?

□ 통화 공급 조절

● 금리 조절

- - 테일러 규칙: 인플레이션과 생산량 갭 (실제 생산량과 잠재 생산량의 차이로, 경제가 가장 효율적일 때 생산할 수 있는 최대 상품 및 서비스 양)

$$i - \pi = r^* + 0.5(\pi - \pi^*) + 0.5(y - y^*)$$

- 여기서 i 는 명목 연방 기금 금리, r^* 는 완전 고용 시의 실질 연방 기금 금리, π 는 인플레이션율, π^* 는 목표 인플레이션율, y 는 실질 생산량의 로그값, y^* 는 완전 고용 시의 잠재 생산량의 로그값이다.
- 테일러 규칙은 인플레이션이 높거나 고용이 완전 고용 수준을 초과할 때 연방 준비제도가 금리를 인상해야 한다고 권장한다. 반대로, 인플레이션과 고용 수준이 낮을 때는 금리를 인하해야 한다.

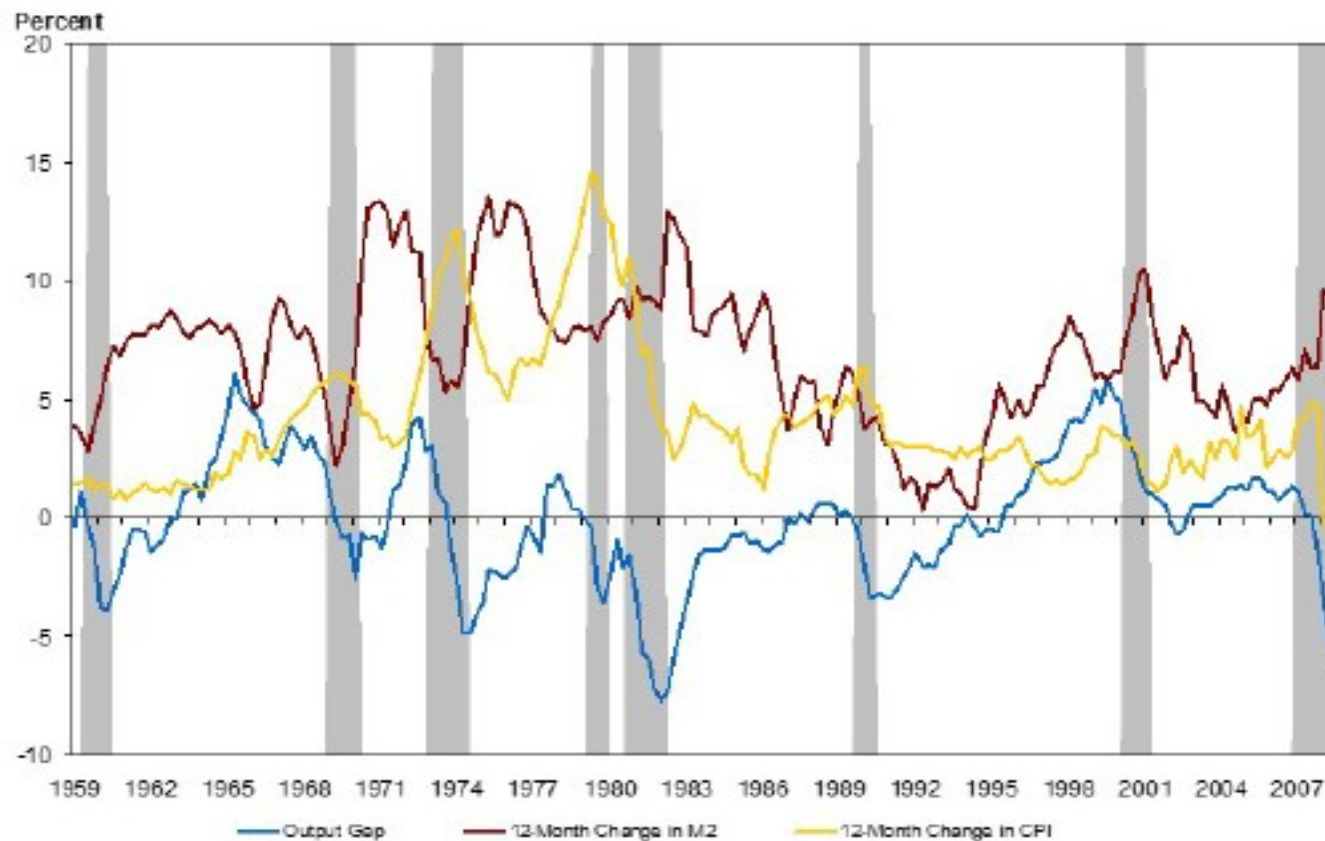
● 공개 시장 운영

- 양적 완화.

통화 정책

□ 통화정책

12-month change in M2, 12-month change in CPI, and estimated output gap: 1959:Q4-2009:Q2

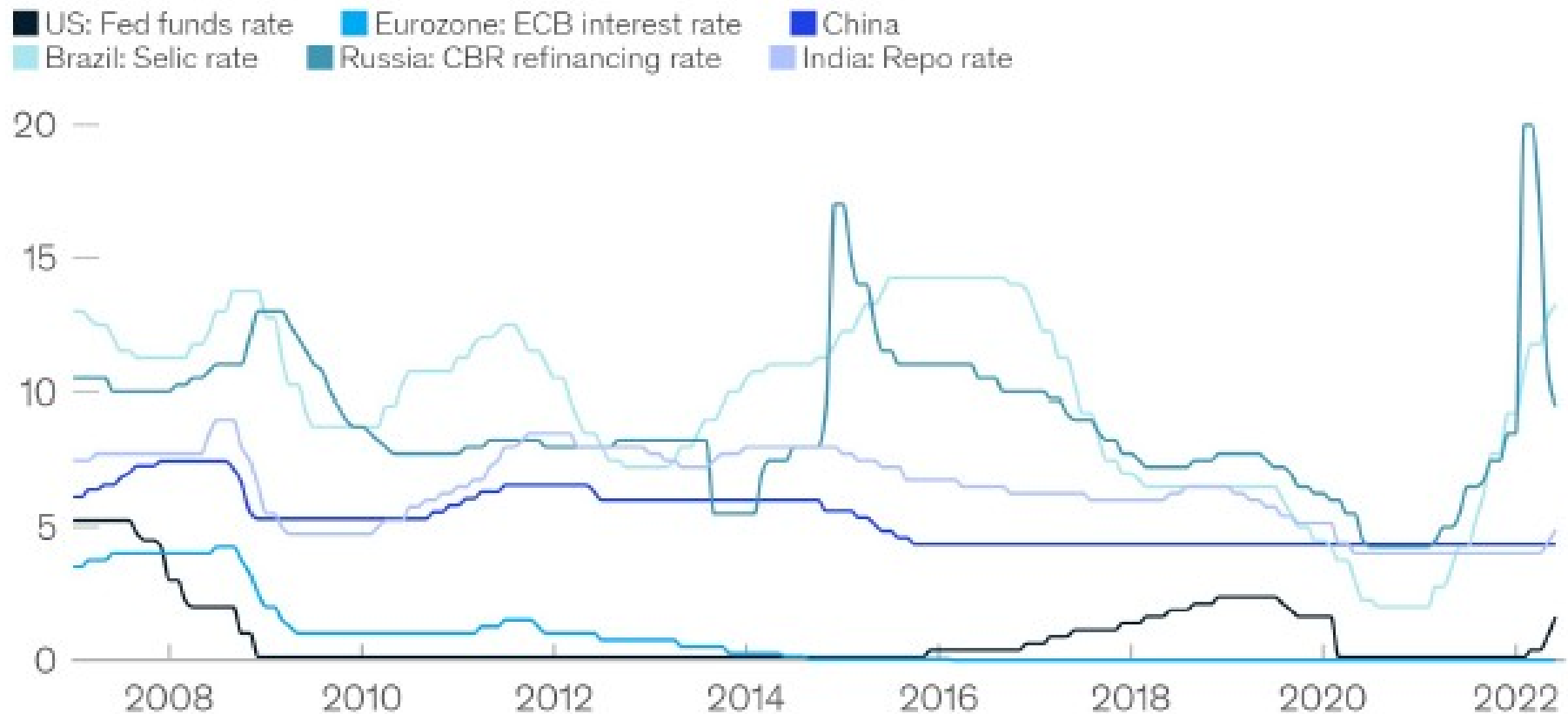


M2 includes M1 plus savings deposits, money market securities, mutual funds, and other time deposits, while M1 includes cash and checking deposits.

최근 인플레이션 상승에 대한 중앙은행의 반응

Most central banks are on a tightening course, attempting to control inflation by raising policy interest rates.

Central bank interest rates, %



Source: Central Bank of Brazil; Central Bank of the Russian Federation; European Central Bank; Federal Reserve Bank of New York; Reserve Bank of India; McKinsey's Global Economics Intelligence analysis

TIPS

- 미국 재무부는 1997년 1월부터 인플레이션 연동 채권(U.S. Treasury Inflation-Protected Securities; TIPS)을 발행하기 시작했다.
- 일반적인 재무부 채권과 달리, TIPS의 원금과 이자 지급은 미국 도시 소비자 물가 지수에 연동된다.
- 이자는 인플레이션 조정된 원금에 고정된 금리로 연간 두 번 지급된다. 디플레이션이 발생하면 원금과 이자 지급액이 하향 조정될 수 있다. 그러나 만기 시, 투자자는 인플레이션 조정된 원금 또는 원래 원금 중 더 큰 금액을 받게 된다.

TIPS 쿠폰 계산

□ TIPS 쿠폰과 원금 지급 계산

Original Principal (par)	\$100,000
Coupon Rate ("Real Yield"), 10Y	1.00%
Nominal Treasury Yield, 10Y	3.00%
Breakeven inflation *	2.00%

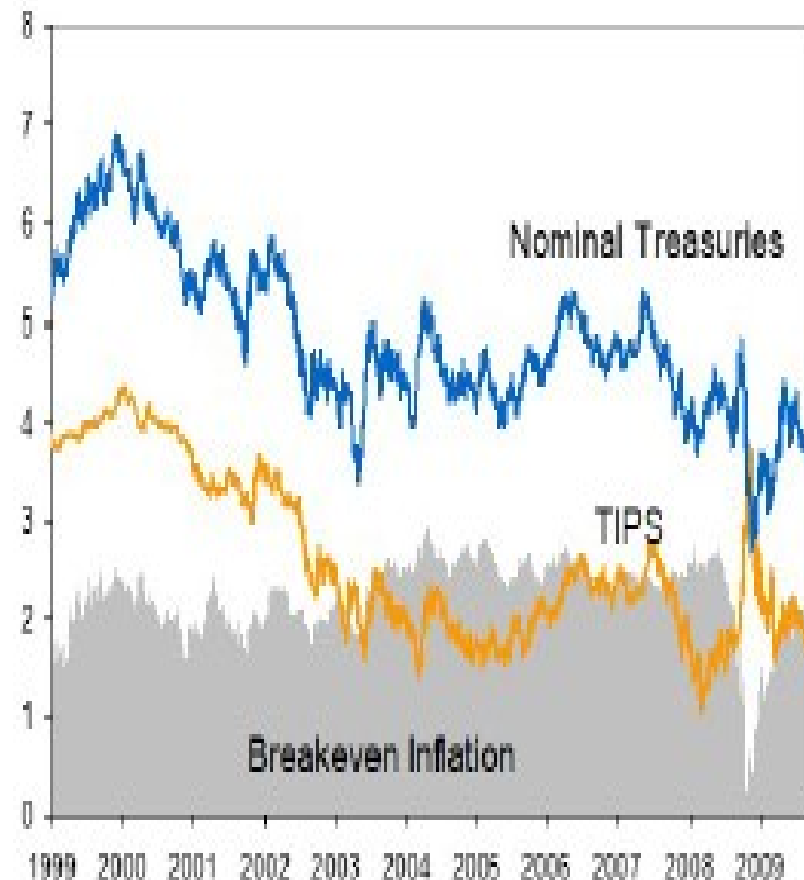
Nominal yield =
Real +
Inflation +
Risk premium

Time (years)	Inflation		Principal	Coupon		
	(annual)	(semi-A)		(annual)	(semi-A)	(\$)
0.0			\$100,000			
0.5	3.0%	1.5%	101,500	1.00%	0.50%	\$508
1.0	4.0%	2.0%	103,530	1.00%	0.50%	518
1.5	1.0%	0.5%	104,048	1.00%	0.50%	520
...						
10.0			MAX[P, \$100,000]			

시장 기대: 과거 10년 TIPS 스프레드

연속 복리 10년 무이표 TIPS 및 재무부 채권 수익률: 1999년 1월 4일 - 2009년 10월 7일

- 왜 TIPS의 수익률이 동일한 만기의 일반 재무부 채권 수익률과 다른가?
 - - 인플레이션 기대
 - - 인플레이션 위험 프리미엄
 - + 유동성 프리미엄
- TIPS 스프레드는 역사적 범위 내에 있습니다:
 - 2009년 10월 7일: 1.99%
 - 샘플 평균: 2.23%



- 손익분기(Breakeven) 인플레이션=일반 재무부 채권 수익률-TIPS 수익률

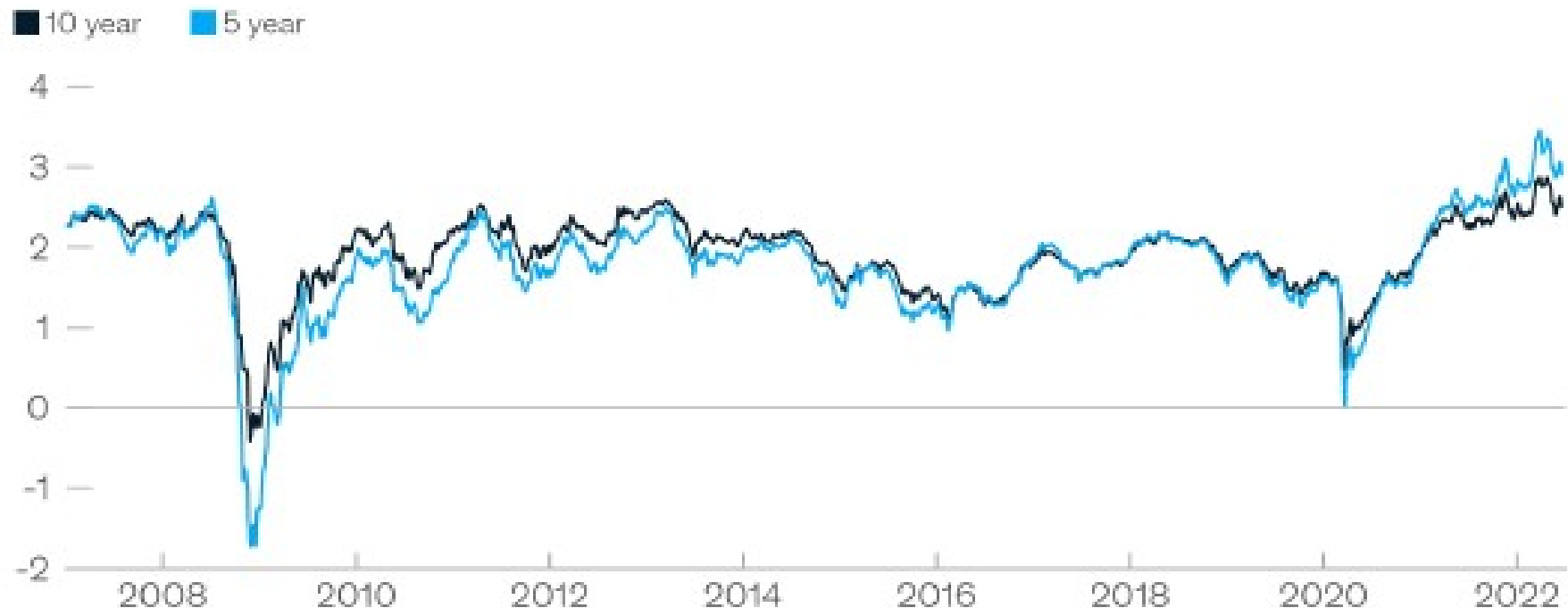
Data from the Federal Reserve Board Finance and Economics Discussion Series. Updated work by Refet S. Gurkaynak, Brian Sack, and Jonathan H. Wright.

<https://www.federalreserve.gov/pubs/feds/2008/200805/200805abs.html>.

미국 TIPS의 수익률 스프레드에 의해 나타난 내재 인플레이션 기대: 5년 대 10년

Inflation expectations eased slightly, as indicated by the yield spread in US Treasury products.

Implied inflationary expectations from 5- and 10-year TIPS yields (spread between T-bill and TIPS of same maturity), 5-day moving average, %



Note: Updated through Jun 17, 2022.

Source: Haver Analytics; McKinsey Global Economics Intelligence analysis

Global Economics Intelligence executive summary, McKinsey Company, June 2022

여러 인플레이션 예측

- 전문가 예측 조사, 2009년 3분기
 - 향후 5년: 2.2%
 - 향후 10년: 2.5%

- 리빙스턴 조사, 2009년 6월
 - 2008년에서 2009년: -0.7%
 - 2009년에서 2010년: 1.7%

- 미시간 대학교, 소비자 인플레이션 기대 조사, 2009년 8월
 - 향후 12개월: 2.82%

Results from the Survey of Professional Forecasters is made available by the Federal Reserve Bank of Philadelphia.
<https://www.philadelphiafed.org/search-results?pageStart=1&pageSize=12&searchQuery=Survey+of+Professional+Forecasters> Livingston Survey forecast from:
<https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/livingston-survey>
University of Michigan Survey of Consumers Inflation Expectation data from
<http://research.stlouisfed.org/fred2/series/MICH/>.

왜 인플레이션을 신경 써야 하는가?

- 구매력에 영향을 미친다.
- 인플레이션이 구매력에 미치는 영향을 어떻게 최소화할 수 있을까?
 - 인플레이션과 함께 가치가 1대1로 증가하는 자산에 투자함으로써 (실질 수익률이 일정하게 유지됨)
- 우리는 단지 인플레이션이 구매력에 미치는 영향을 최소화하는 것에만 집중해야 할까?

인플레이션과 투자 결정: 포트폴리오 접근법

□ 투자자의 목표:

- 실질 수익률의 불확실성을 수용 가능한 수준으로 유지하면서 포트폴리오의 예상 실질 수익률을 최대화하는 것.

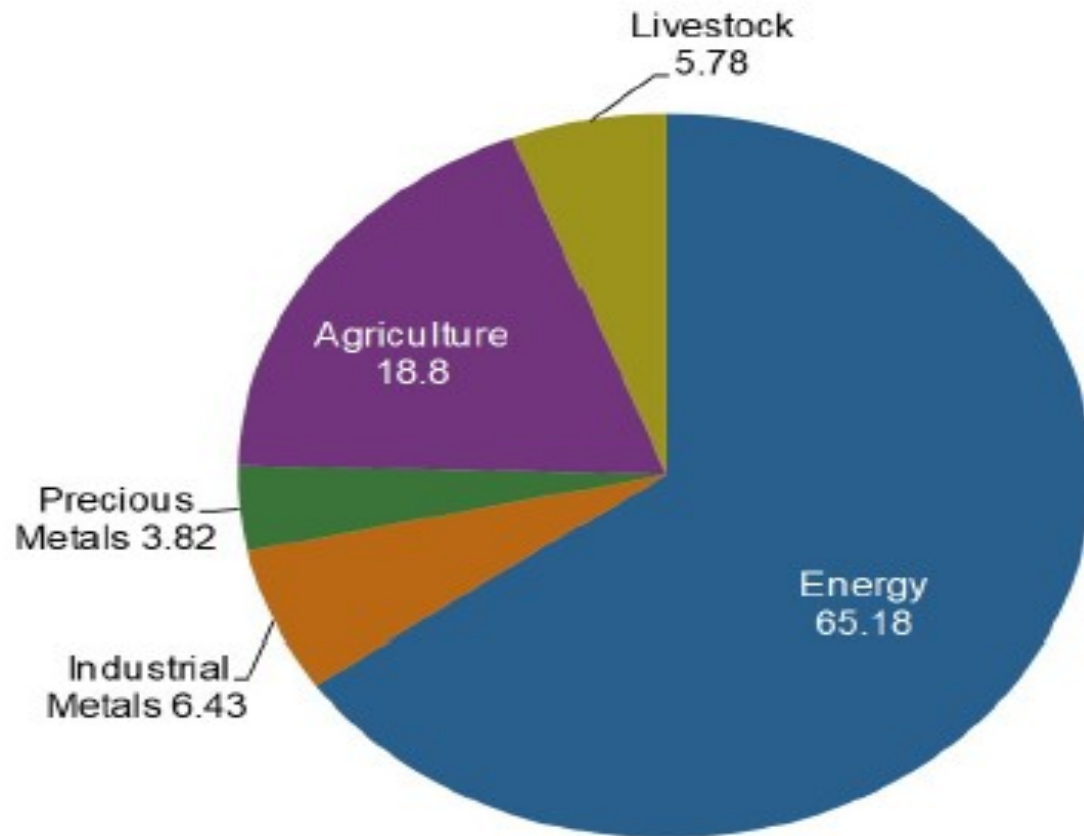
상품(Commodities)

- 상품은 CPI에 포함된 재화 및 서비스 바스켓의 약 40%를 차지한다.
- 상품 가격은 인플레이션 측정에 반영되기 때문에, 상품의 수익률과 인플레이션 사이에는 기계적인 관계가 있다.
- 상품과 인플레이션 간의 양의 상관관계는 많은 사람들에게 상품 선물에 장기 투자하는 것이 인플레이션에 대한 좋은 헤지 수단이라는 결론을 내리게 했다.
- 포트폴리오를 설계할 때 인플레이션과의 상관관계도 고려해야 하지만, 자산이 포트폴리오에 가져오는 이익은 자산의 예상 수익률, 수익률 분산, 포트폴리오 내 다른 자산과의 공분산에도 의존한다.
- 상품은 인플레이션과 유리한 상관관계를 가질 수 있지만, 그들의 위험 및 수익 특성은 덜 매력적으로 만든다.

상품(Commodities)

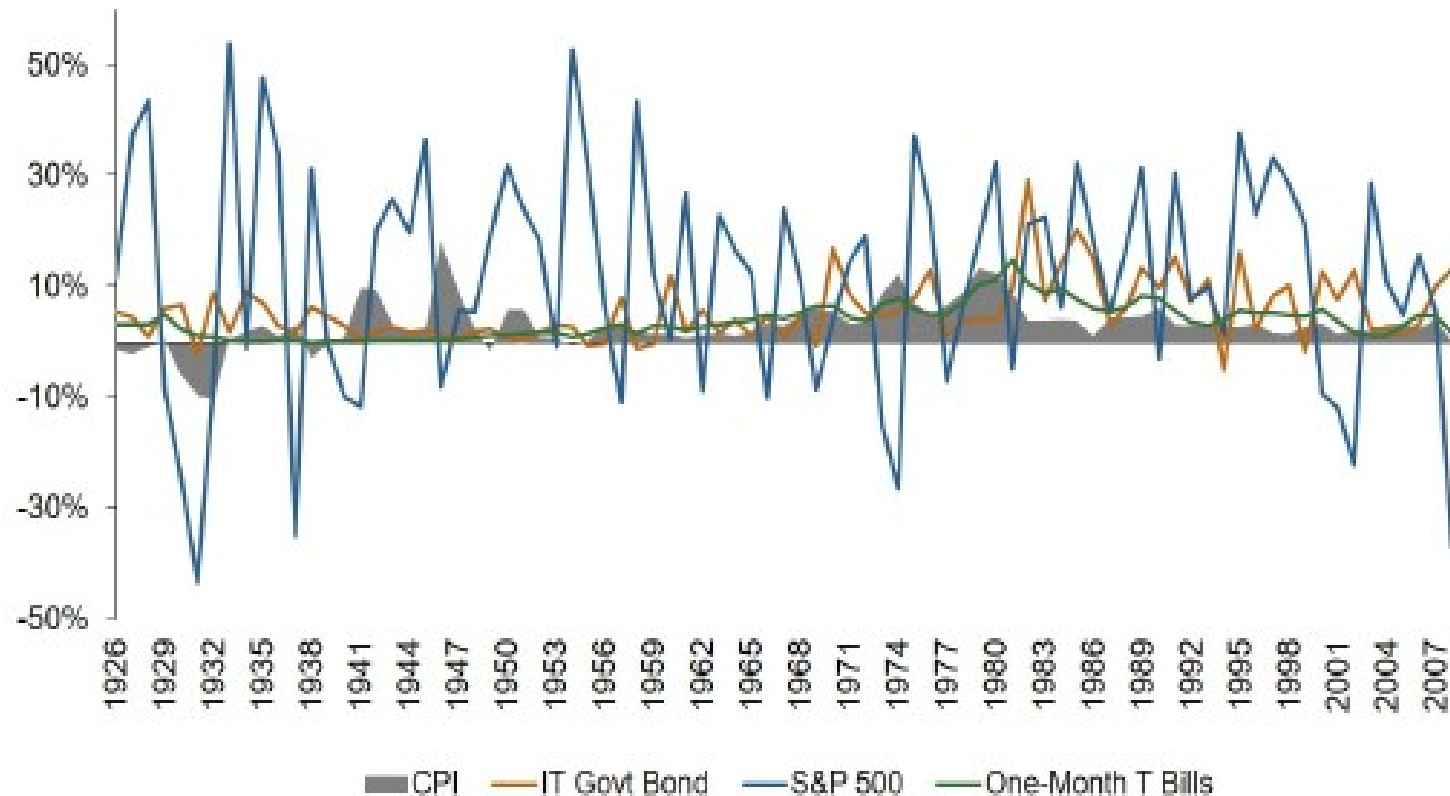
- 장기 투자자들은 포트폴리오가 인플레이션을 얼마나 잘 앞설 수 있는지에 대해 걱정한다.
- 높은 예상 수익률을 가진 포트폴리오가 장기적으로 구매력을 보존하고 증가시킬 최고의 기회를 가질 것이다.
- 상품은 인플레이션과 양의 상관관계를 가지지만, 매우 변동성이 크다.

S&P GSCI



Check [this](#) for 2022 weights.

연간 수익률: 1926 - 2008 (미국)



Lee & Lee, 2009, Dimensional Fund Advisors

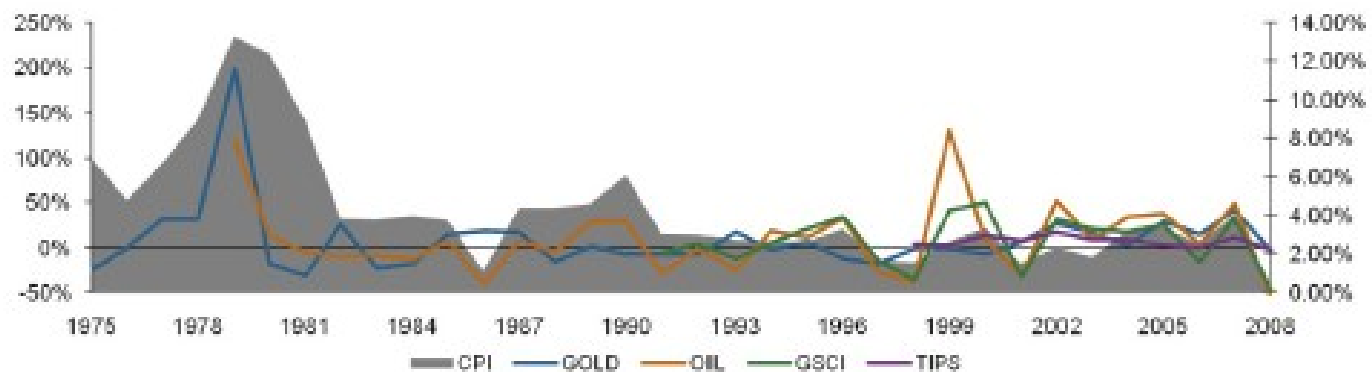
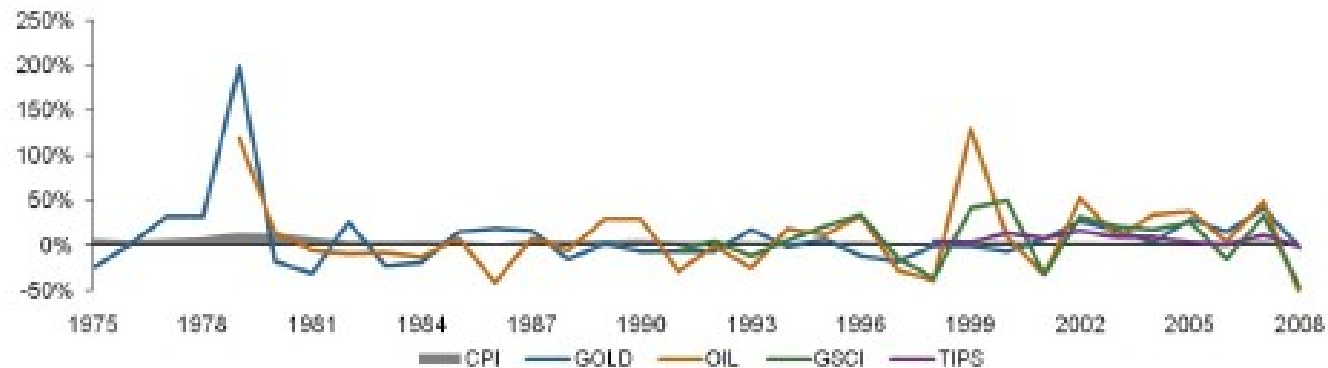
All items CPI-U data from the US Bureau of Labor Statistics.

IT Government Bonds and One-month Treasury bill data from Ibbotson Associates.

The S&P data are provided by Standard & Poor's Index Services Group.

Index is not available for direct investment; its performance does not reflect the expenses associated with the management of an actual portfolio.

연간 수익률: 1926 - 2008 (미국)



Lee & Lee, 2009, Dimensional Fund Advisors

All items CPI-U data from the US Bureau of Labor Statistics.

IT Government Bonds and One-month Treasury bill data from Ibbotson Associates.

The S&P data are provided by Standard & Poor's Index Services Group.

Index is not available for direct investment; its performance does not reflect the expenses associated with the management of an actual portfolio.

회귀 결과: 연간 명목 수익률 (종속 변수) 대비 인플레이션 (독립 변수)

	Inflation	t-stat	Residual Std Dev (%)	# of years
Market Index(CRSP)	0.097	0.13	20.53	83
LTGB	-0.335	-1.82	9.25	83
ITGB	0.000	0.00	5.66	83
T1Y	0.360	3.04	3.73	68
T1M	0.299	2.69	2.81	83
TIPS	2.481	2.52	5.05	11
GSCI	22.487	6.95	18.96	18
Gold	5.196	1.00	34.87	34
Oil	6.246	2.04	36.83	30

Lee & Lee, 2009, Dimensional Fund Advisors

All items' CPI-U data from the US Bureau of Labor Statistics. LT Government Bonds, IT Government Bonds, and One-Month Treasury bill (T1M) data from Ibbotson Associates. One-Year Treasury (T1Y) data are from the CRSP and S&P 500 data are from Standard & Poor's. TIPS data are from Barclay's Capital (Barclays Capital US TIPS index). Commodities data are from Standard & Poor's (S&P GSCI Index). Gold data are obtained from the World Gold Council website (<http://www.research.gold.org/prices/>), London PM Fix gold price. Oil prices (West Texas Intermediate Crude oil prices) are from <http://www.economagic.com/em-cgi/data.exe/var/west-texas-crude-long#Data>. The S & P data are provided by Standard & Poor's Index Services Group.

2008년까지의 연간 수익률에 대한 요약 통계량

	Market Index (CRSP)	LTGB	ITGB	T1Y	T1M	TIPS	GSCI	Gold	Oil	Inflation
Correlation with Inflation	0.020	-0.151	0.000	0.181	0.410	0.483	0.740	0.403	0.443	
Nominal Returns										
Average	11.41	6.09	5.56	5.23	3.75	6.68	6.73	8.86	10.09	3.10
Std. Dev.	20.53	9.36	5.66	3.85	3.08	5.77	28.18	38.10	41.09	4.22
S.E.	2.25	1.03	0.62	0.47	0.34	1.74	6.64	6.53	7.50	0.46
T-stat	5.06	5.93	8.95	11.20	11.09	3.84	1.01	1.36	1.35	6.69
Real Returns										
Average	8.31	2.99	2.46	1.11	0.65	4.23	4.17	4.60	6.20	
Std. Dev.	20.88	10.84	7.07	4.73	4.08	5.32	27.50	37.01	39.89	
S.E.	2.29	1.19	0.78	0.57	0.45	1.60	6.48	6.35	7.28	
T-stat	3.63	2.52	3.18	1.93	1.44	2.64	0.64	0.73	0.85	
Sample Start Date	1926	1926	1926	1941	1926	1998	1991	1975	1979	1926

Lee & Lee, 2009, Dimensional Fund Advisors

All items' CPI-U data from the US Bureau of Labor Statistics. LT Government Bonds, IT Government Bonds, and One-Month Treasury bill (T1M) data from Ibbotson Associates. One-Year Treasury (T1Y) data are from the CRSP and S& P 500 data are from Standard & Poor's. TIPS data are from Barclay's Capital (Barclays Capital US TIPS index). Commodities data are from Standard & Poor's (S& P GSCI Index). Gold data are obtained from the World Gold Council website (<http://www.research.gold.org/prices/>), London PM Fix gold price. Oil prices (West Texas Intermediate Crude oil prices) are from

<http://www.economagic.com/em-cgi/data.exe/var/west-texas-crude-long#Data> The S & P data are provided by Standard & Poor's Index Services Group.

역사적 증거

- 과거에 상품, 금, 석유 또는 TIPS가 실질 수익률 변동성을 줄이는 데 사용될 수 있었을까?
- 인플레이션 헤징 자산을 주식과 채권으로 구성된 기본 포트폴리오에 추가한다.
- 실험에서는 연간 리밸런싱을 사용한다.

연간 수익률 변동성의 변화: 설명용 포트폴리오

Portfolio Weights		Portfolio Real Return Standard Deviation			
Base	Asset	TIPS	GSCI	Gold	Oil
Base Portfolio: 80% CRSP 1-10, 10% ITGB, 10% T1M					
100	0	16.43	16.04	14.17	14.16
90	10	14.67	14.94	12.92	13.45
80	20	12.94	14.42	12.90	14.02
70	30	11.25	14.54	14.13	15.73
Base Portfolio: 60% CRSP 1-10, 20% ITGB, 20% T1M					
100	0	11.82	12.03	10.90	10.84
90	10	10.54	11.35	10.04	10.41
80	20	9.29	11.40	10.71	11.58
70	30	8.08	12.17	12.00	13.95
Base Portfolio: 40% CRSP 1-10, 30% ITGB, 30% T1M					
100	0	7.30	8.15	7.87	7.76
90	10	6.48	7.87	7.42	7.53
80	20	5.72	8.58	8.88	9.38
70	30	5.04	10.09	11.54	12.41
Starting Year		1998	1991	1975	1979

Lee & Lee, 2009, Dimensional Fund Advisors

Real returns are estimated using all items' CPI-U data from the US Bureau of Labor Statistics. LT Government Bonds, IT Government Bonds, and One-Month Treasury bill (T1M) data from Ibbotson Associates. One-Year Treasury (T1Y) data are from the CRSP and S& P 500 data are from Standard & Poor's. TIPS data are from Barclay's Capital (Barclays Capital US TIPS index). Commodities data are from Standard & Poor's (S& P GSCI Index). Gold data are obtained from the World Gold Council website (<http://www.research.gold.org/prices/>), London PM Fix gold price. Oil prices (West Texas Intermediate Crude oil prices) are from <http://www.economagic.com/em-cgi/data.exe/var/west-texas-crude-long#Data> The S & P data are provided by Standard & Poor's Index Services Group.

인플레이션 방어 포트폴리오를 구축하는 방법?

- 인플레이션 방어 포트폴리오를 구축할 때 목표는 적절한 수준의 위험을 유지하면서 포트폴리오의 구매력을 최대화하는 것이다.
- 위험이 실질 수익률의 변동성으로 측정된다면, 매력적인 자산은 다음과 같은 특성을 가져야 한다:
 - 높은 예상 실질 수익률
 - 낮은 실질 수익률 변동성
 - 다른 자산과의 낮은 상관관계

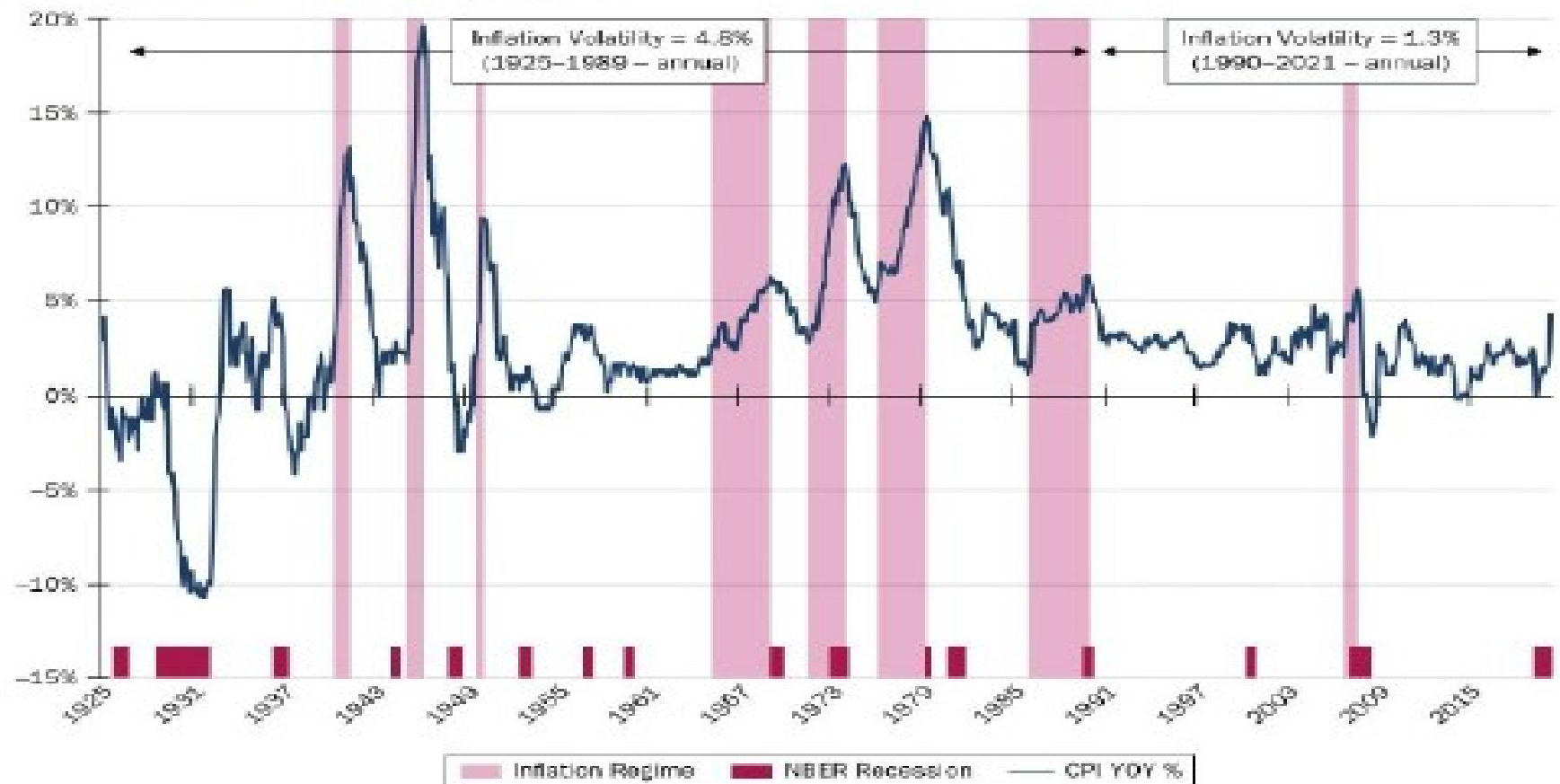
□ 인플레이션 시기에 가장 좋은 전략

- Neville, et al.(2021, JPM)는 1926년부터 2021년 4월까지 미국, 영국, 일본에서 인플레이션 시기에 어떤 투자 전략이 더 나은 성과를 내는지 조사했다.
- 이 논문은 패시브 주식 및 채권 전략이 인플레이션 급등 시기에 저조한 성과를 보인다는 것을 보여준다.
- 상품, 미술품 및 와인과 같은 수집품, 그리고 추세 추종 및 우량주 팩터(quality factor) 기반의 주식 전략이 인플레이션 시기에 더 나은 성과를 보인다..

인플레이션, 인플레이션 변동성과 경기 침체

□ Neville et al. (2021, JPM)

EXHIBIT 1
US YoY CPI Overlaid with Inflationary Regimes



NOTES: The percentage increase in the US headline Consumer Price Index (CPI) basket over the trailing 12-month period. The pink shaded areas demarcate our quantitatively defined inflationary regimes, which are described in more detail later. The red rectangles at the bottom of the chart are periods when the US economy was in recession, as defined by the National Bureau of Economic Research (NBER). The data are monthly, from 1926 to April 2021. See Appendix A for more details.

Neville, et. al. (2021, JPM)

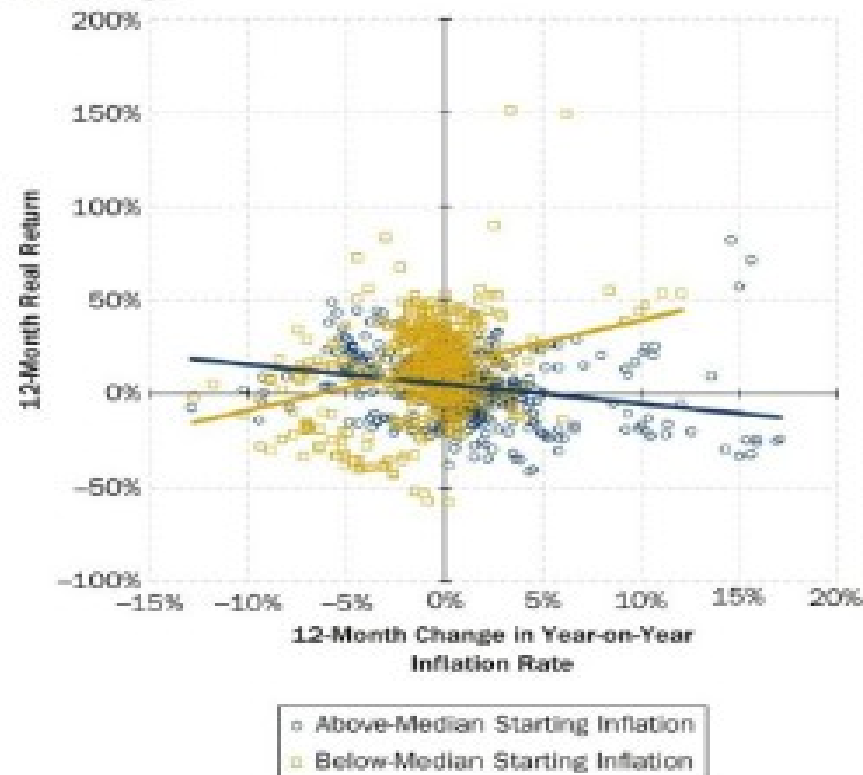
실질 주식 수익률 대 같은 시기의 12개월 인플레이션 변화

□ Neville et al. (2021, JPM)

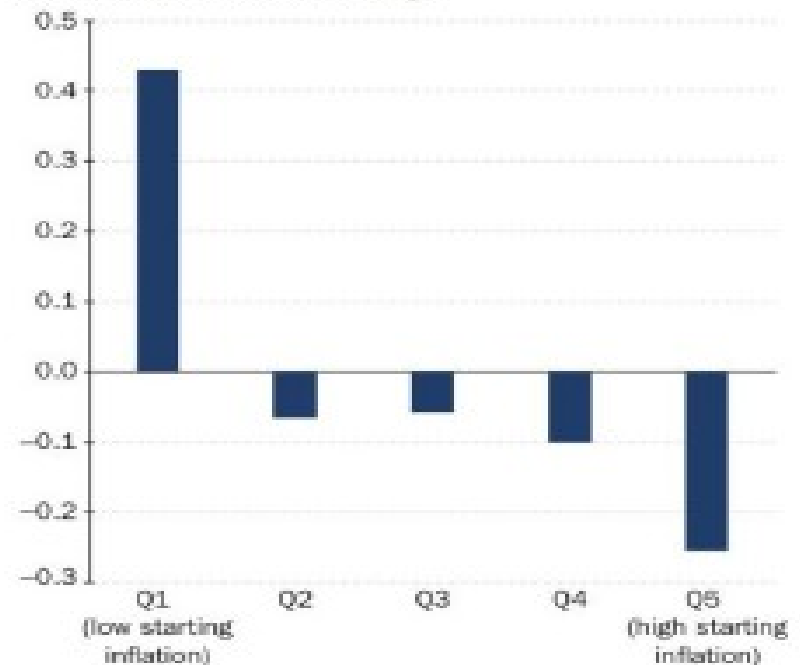
EXHIBIT 5

Real US Equity Return versus Contemporaneous 12-Month Inflation Rate Change

Panel A: 12-Month Real Equity Return vs. Inflation Rate Change



Panel B: Correlation of 12-Month Real Equity Returns and Inflation Rate Change



NOTES: The relation between the 12-month real US equity return and the contemporaneous 12-month change in the YoY inflation rate. Panel A plots the data, splitting the data points into the above- and below-median inflation rate at the start of the 12-month evaluating window, and we add a trend line for both cases. Panel B shows the correlation, splitting the data points into five quintiles based on the inflation rate at the start of the 12-month evaluation window. We use all overlapping 12 months in the 1926–2020 sample period.

Neville, et. al. (2021, JPM)

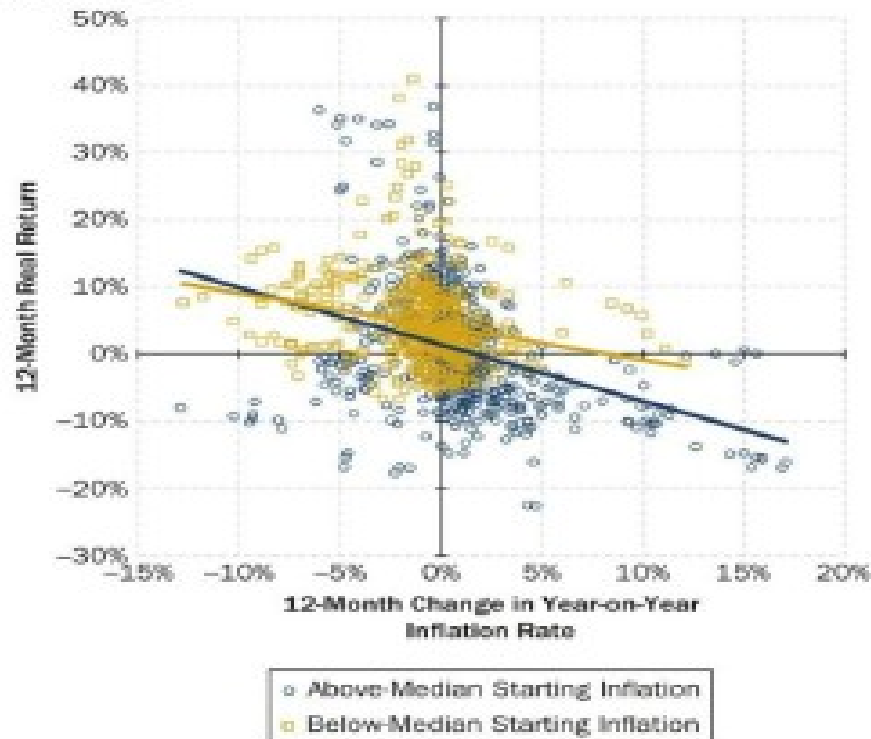
실질 채권 수익률 대 같은 시기의 12개월 인플레이션 변화

□ Neville et al. (2021, JPM)

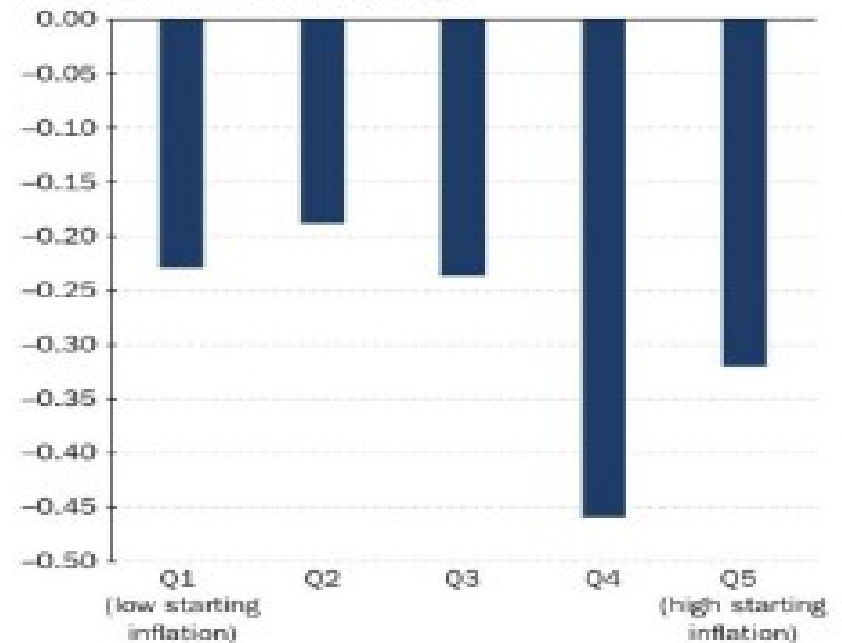
EXHIBIT 6

Real US Bond Return vs. Contemporaneous 12-Month Inflation Rate Change

Panel A: 12-Month Real Bond Return vs. Inflation Rate Change



Panel B: Correlation of 12-Month Real Bond Returns and Inflation Rate Change



NOTES: The relationship between the 12-month real US 10-year Treasury return and the contemporaneous 12-month change in the YoY inflation rate. Panel A plots the data, splitting the data points into the above- and below-median inflation rate at the start of the 12-month evaluation window, and we add a trend line for both cases. Panel B shows the correlation, splitting the data points into five quintiles based on the inflation rate at the start of the 12-month evaluation window. We use all overlapping 12 months in the 1928–2020 sample period.

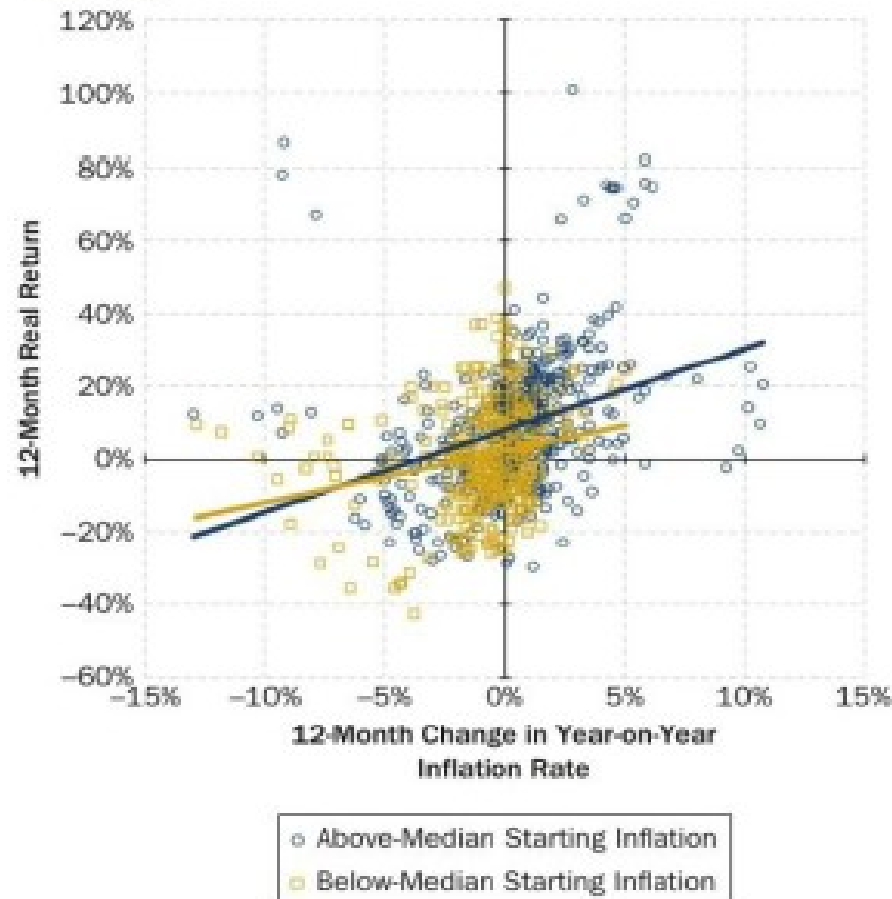
실질 상품 수익률 대 같은 시기의 12개월 인플레이션 변화

□ Neville et al. (2021, JPM)

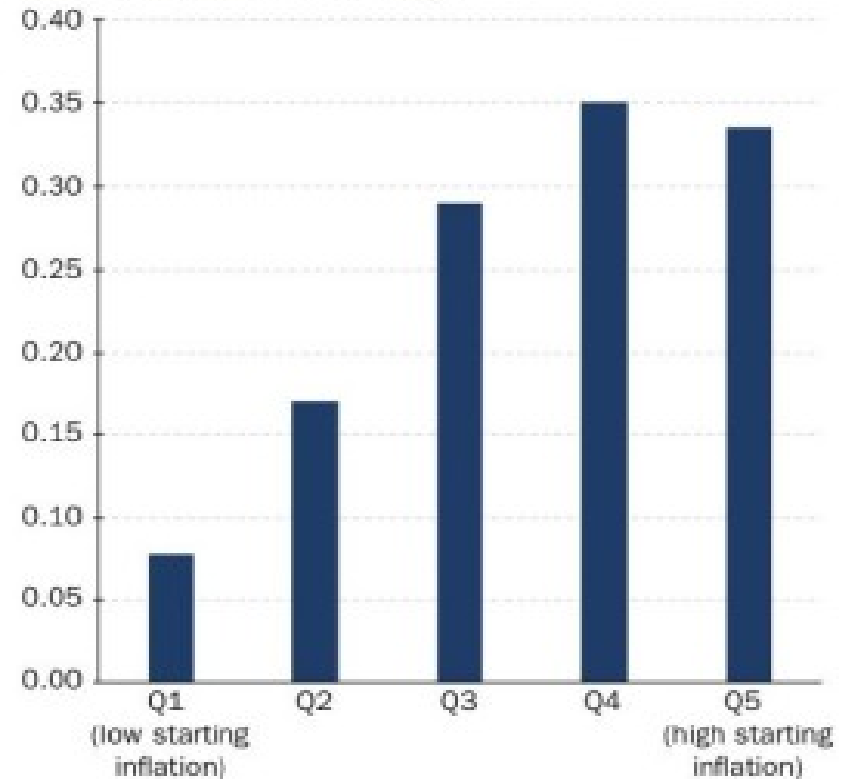
EXHIBIT 10

Real Commodity Returns vs. Contemporaneous 12-Month Inflation Rate Change

Panel A: 12-Month Real Commodities Return vs. Inflation Rate Change



Panel B: Correlation of 12-Month Real Commodity Returns and Inflation Rate Change



Neville, et. al. (2021, JPM)

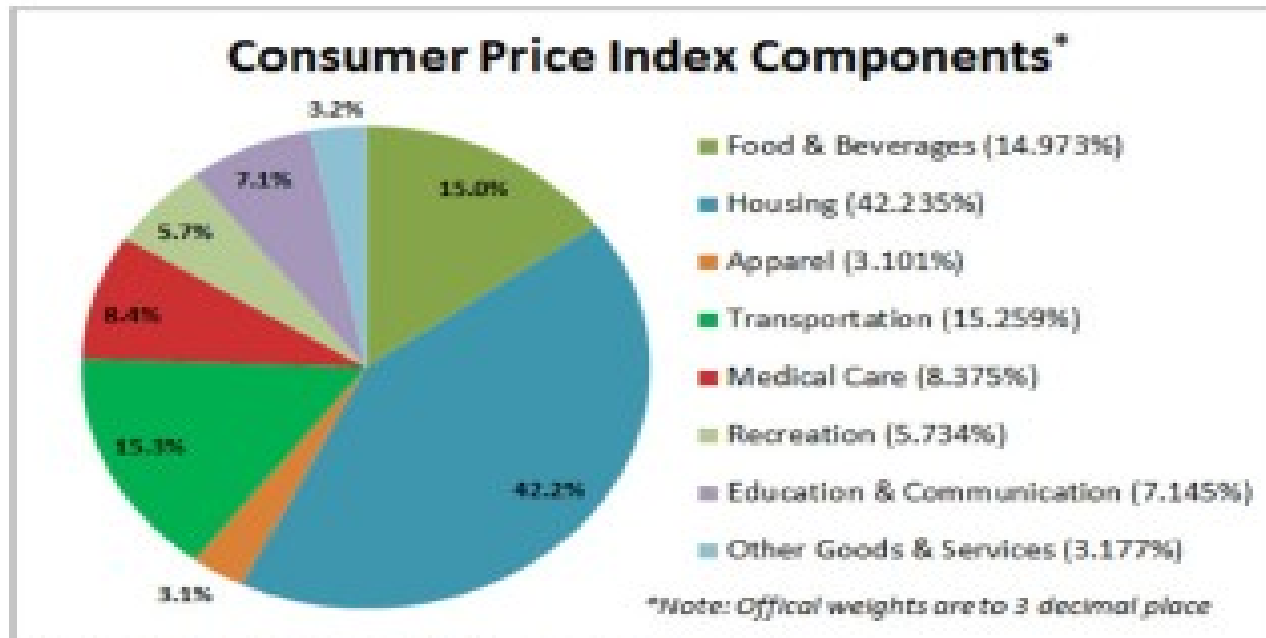
부록: 인플레이션과 실물 자산

인플레이션의 측정

- 소비자 물가 지수(CPI)의 변화
 - All Items, All Urban consumers: TIPS에 사용됨
 - CORE (식품과 에너지는 포함되지 않음)
- 생산자 물가 지수(PPI)
- 개인 소비 지출(PCE)
- GDP 디플레이터

CPI-All Urban Consumers

- 주택 가격은 자가 소유자의 주거에 대한 임대료 상당액을 통해 반영된다.
- 주택에는 임대 비용, 연료, 공과금, 가구 및 유지 보수 관련 비용이 포함된다.



Source: BLS; The most recent reweighting was in December 2015.

From <https://www.businessinsider.com/consumer-price-inflation-components-2014-6>

TIP를 통한 기대 인플레이션 추정치

□ 유동성 프리미엄 추정치

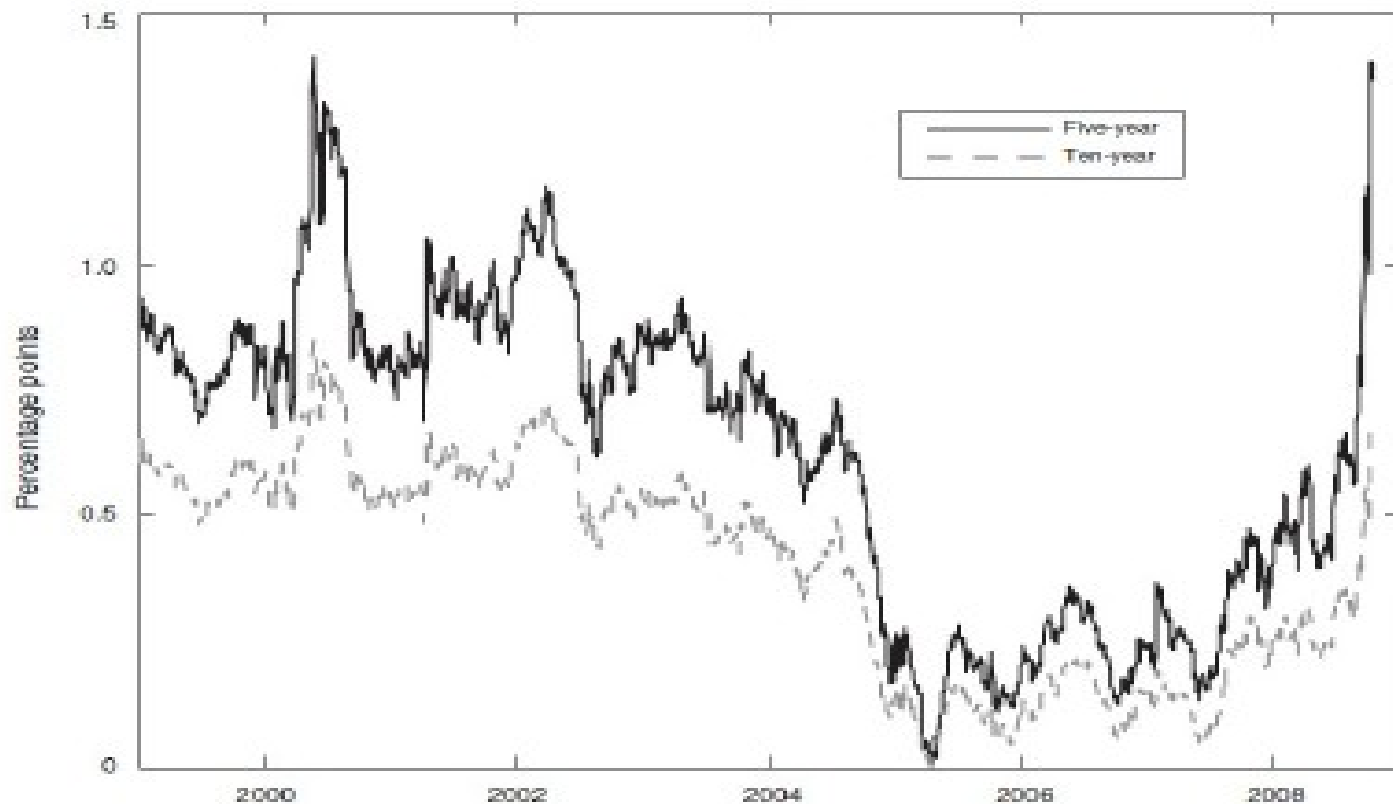


FIGURE 9. ESTIMATED TIPS LIQUIDITY PREMIUM

Notes: Regression-based estimates of the liquidity premium for holding TIPS, relative to that for nominal securities, constructed as described in the text, normalized to zero in April 2005. The regression coefficient estimates are shown in Table 3.

Gurkaynak, Sack and Wright, *American Economic Journal: Macroeconomics*, 2010

TIP를 통한 기대 인플레이션 추정치

□ 유동성 프리미엄 조정 기대 인플레이션

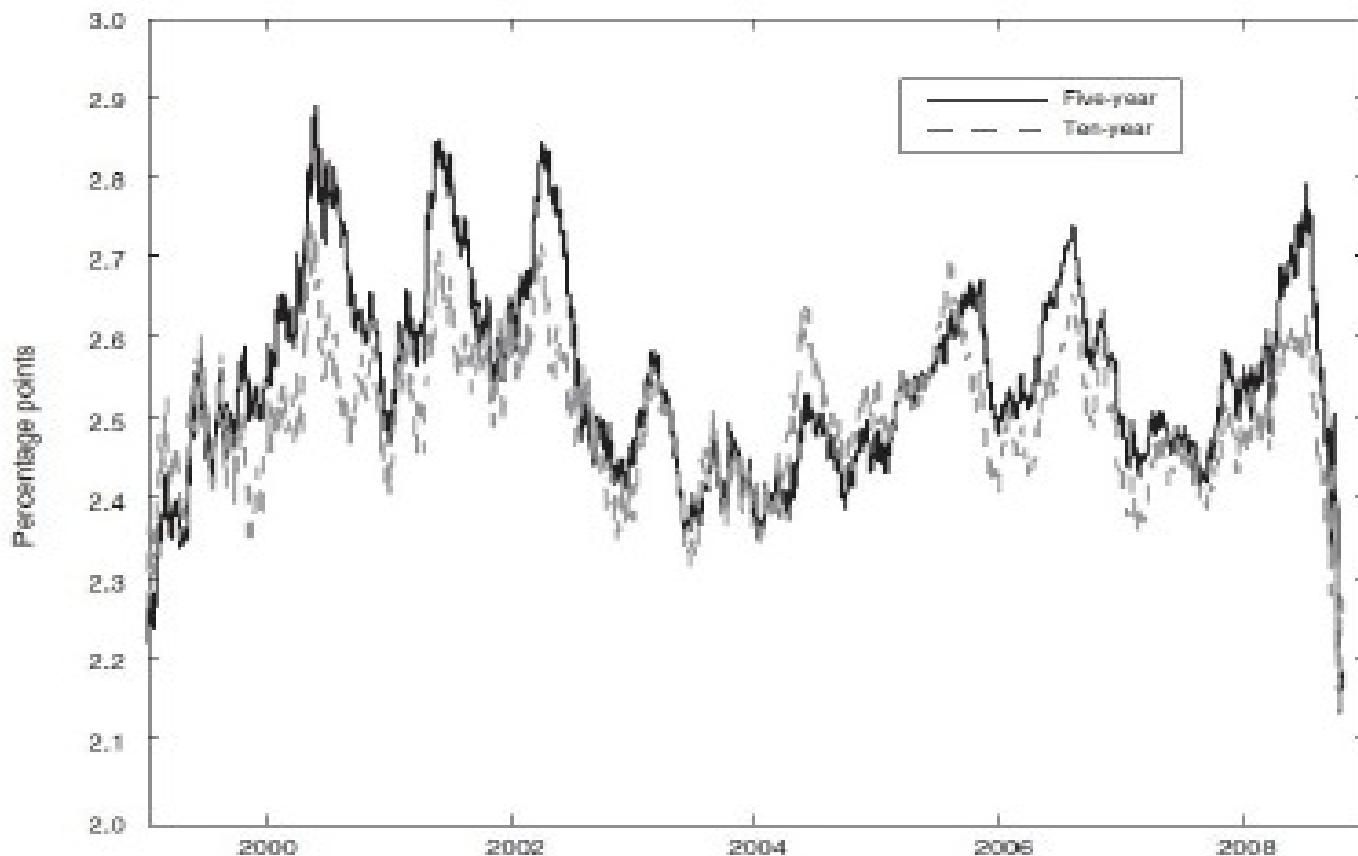


FIGURE 10. ESTIMATED INFLATION EXPECTATIONS

Note: Estimates of inflation expectations at five- and ten-year horizons obtained from the Kalman smoother applied to the inflation compensation series adjusted for liquidity effects.

Gurkaynak, Sack and Wright, American Economic Journal: Macroeconomics, 2010

TIPS

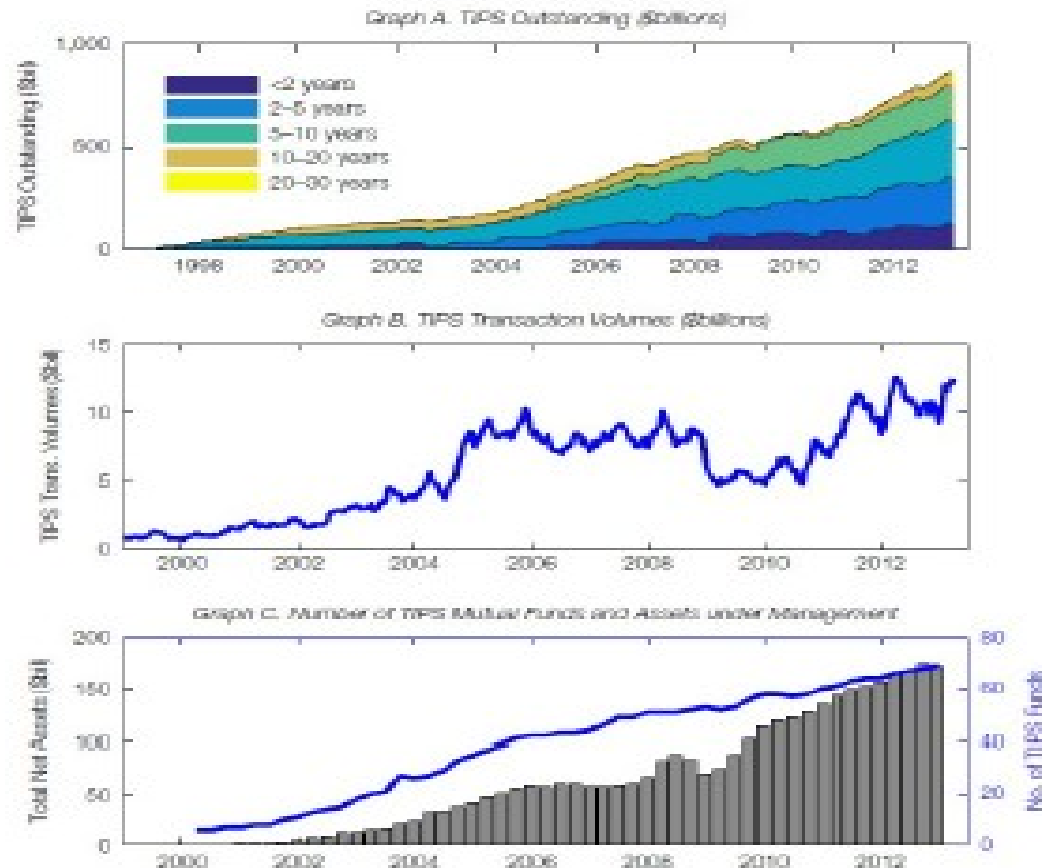
- D'Amico, Kim, Wei (2018, JFQA)는 초기 몇 년 동안 TIPS 수익률에 약 1%의 유동성 위험 프리미엄이 포함되어 있다고 추정했다. 시장이 발전하고 유동성 프리미엄이 감소하면서 TIPS는 매우 큰 수익을 기록했으며, 이는 앞으로 다시 발생할 것으로 기대해서는 안 된다.
- 설계상, TIPS는 투자자를 인플레이션 위험으로부터 보호한다. 인플레이션 위험에 대해 헤징되어 있기 때문에, TIPS 수익률에는 인플레이션 위험 프리미엄이 포함되지 않아야 한다. 장기적으로 TIPS 투자자는 비슷한 만기의 명목 재무부 채권 투자자보다 낮은 수익을 기대해야 한다. 사실, 정부의 차입 비용을 줄이는 것이 재무부가 TIPS를 발행하는 동기 중 하나로 자주 언급된다.

TIPS 시장 규모

□ 시장규모

FIGURE 1
TIPS Outstanding, Transaction Volumes, and Mutual Funds

Graph A of Figure 1 plots TIPS outstanding broken down by remaining maturities, based on data reported in the Treasury's Monthly Statement of the Public Debt (MSPD). Graph B plots the weekly TIPS transaction volumes, defined as the 13-week moving average of weekly averages of daily TIPS transaction volumes reported by primary dealers in Government Securities Dealers Reports (FR-2004). Graph C plots the number of TIPS mutual funds (right axis) and the total net assets under management (left axis) from the Investment Company Institute (<http://www.ici.org>).



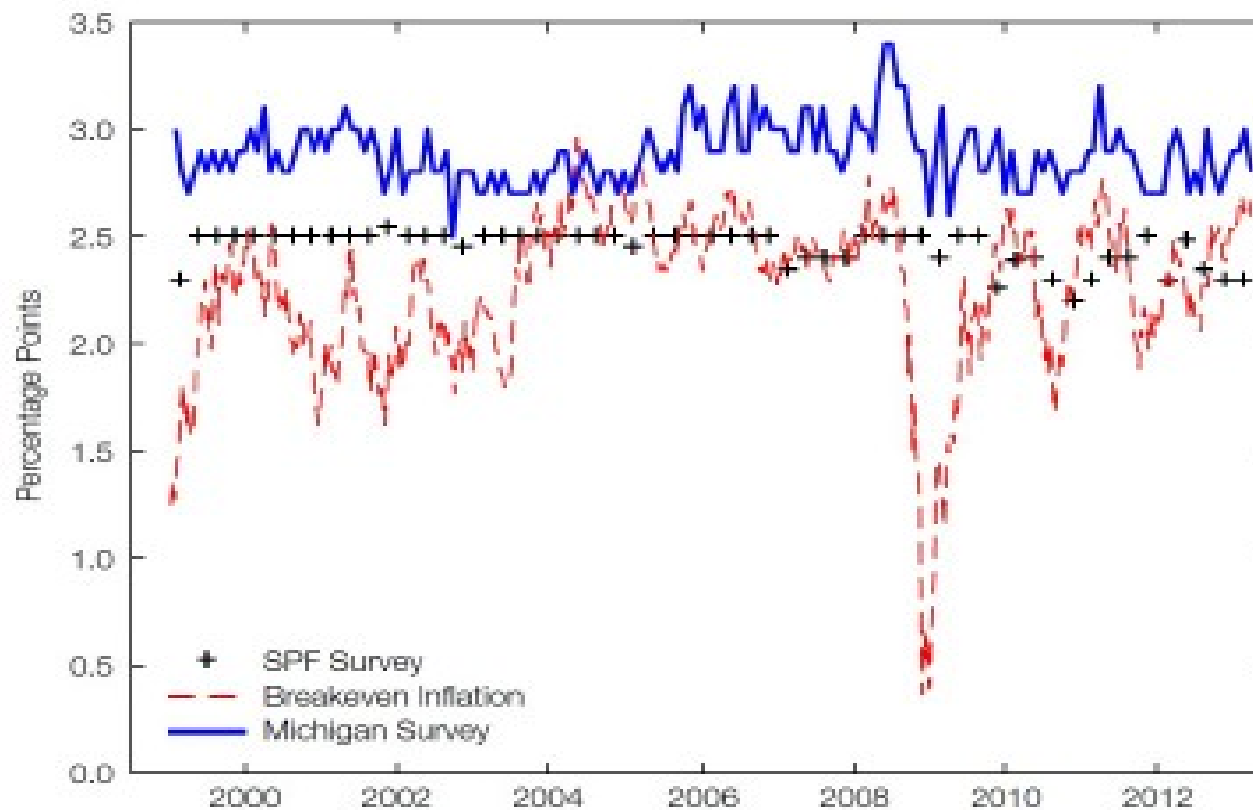
D'Amico, Kim and Wei (2018, JFQA)

TIPS BEI와 인플레이션 예측

□ 서베이 인플레이션 예측과 TIPS BEI

FIGURE 2
Survey Inflation Forecasts and TIPS BEI

Figure 2 shows the 10-year TIPS break-even inflation (BEI) (dashed line), long-horizon Michigan inflation forecast (solid line), and 10-year Survey of Professional Forecasters (SPF) inflation forecast (pluses).



D'Amico, Kim and Wei (2018, JFQA)

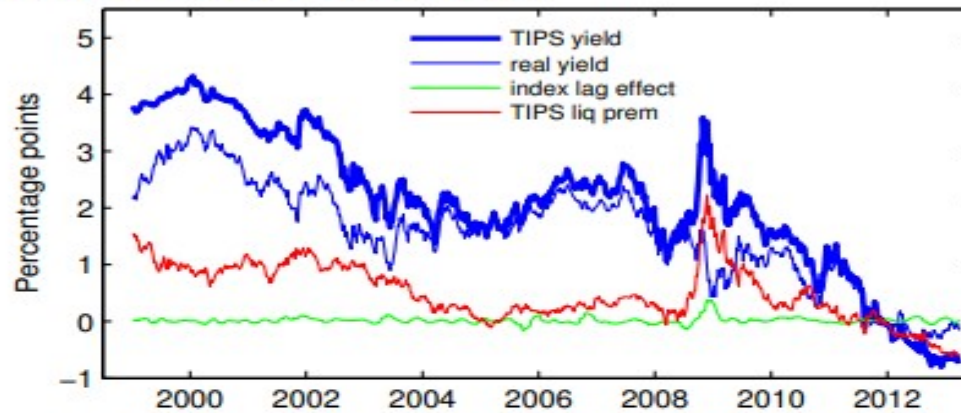
TIPS

□ TIPS 수익률과 TIPS BEI의 분해

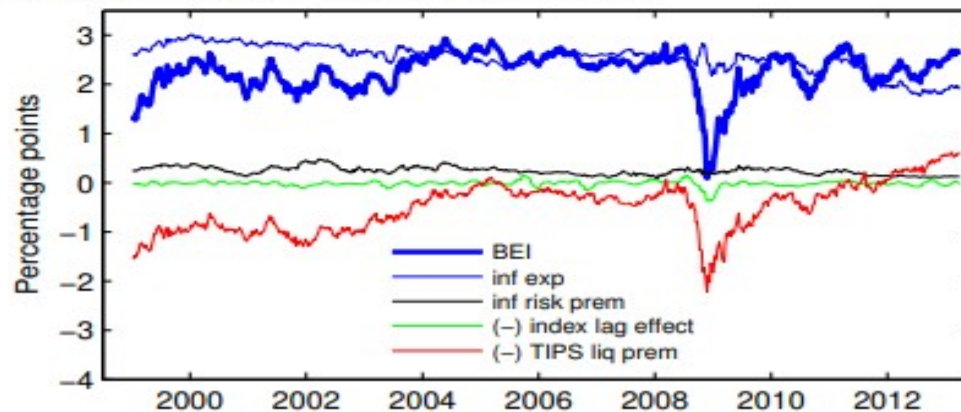
Figure 10: Decomposing TIPS Yields and TIPS Breakeven Inflation

Graph A of Figure 10 decomposes the 10-year TIPS yield into the real yield and the TIPS liquidity premium, while Graph B decomposes the 10-year TIPS BEI into expected inflation, the inflation risk premium and the TIPS liquidity premium, both according to equation (30).

Graph A. Decomposing 10-Year TIPS Yields



Graph B. Decomposing 10-Year Breakeven Inflation



연간 실질 수익률의 상관관계: 가능한 모든 데이터를 사용한 쌍별 상관관계

□ 상관관계

	CRSP 1-10	LTGB	ITGB	T1Y	T1M	TIPS	GSCI	Gold	Oil
CRSP 1-10	1.00								
LTGB	0.13	1.00							
ITGB	0.10	0.93	1.00						
T1Y	0.26	0.71	0.86	1.00					
T1M	0.12	0.55	0.71	0.94	1.00				
TIPS	-0.23	0.23	0.36	-0.04	-0.31	1.00			
GSCI	0.11	-0.25	-0.32	-0.43	-0.33	0.48	1.00		
Gold	-0.08	-0.12	-0.18	-0.35	-0.41	0.24	0.20	1.00	
Oil	0.06	-0.47	-0.51	-0.52	-0.47	0.13	0.80	0.50	1.00

Lee & Lee, 2009, Dimensional Fund Advisors

All items' CPI-U data from the US Bureau of Labor Statistics. LT Government Bonds, IT Government Bonds, and One-Month Treasury bill (T1M) data from Ibbotson Associates. One-Year Treasury (T1Y) data are from the CRSP and S& P 500 data are from Standard & Poor's. TIPS data are from Barclay's Capital (Barclays Capital US TIPS index). Commodities data are from Standard & Poor's (S& P GSCI Index). Gold data are obtained from the World Gold Council website (<http://www.research.gold.org/prices/>), London PM Fix gold price. Oil prices (West Texas Intermediate Crude oil prices) are from <http://www.economagic.com/em-cgi/data.exe/var/west-texas-crude-long#Data> The S & P data are provided by Standard & Poor's Index Services Group.

미국 인플레이션 국면별 성과

□ 미국 인플레이션 국면별 성과

EXHIBIT 2

Summary Performance in US Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes		
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)
Characteristics											
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007			
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008			
Length (months)	14	13	7	48	30	38	46	11			
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%			
Strategy	Real Return (total)								Real Return (annualized)		
(P) Commodities—Energies	-3%	2%	-6%	-16%	264%	57%	201%	68%	41%	-1%	3%
(A) Trend—All Assets	20%	23%	19%	135%	196%	100%	65%	17%	25%	15%	16%
(A) Trend—Commodities			1%	54%	173%	33%	132%	25%	20%	8%	10%
(P) Commodities—Industrial				115%	38%	-6%	306%	3%	19%	4%	7%
(A) Trend—Bonds				79%	54%	149%	6%	6%	15%	9%	10%
(P) Commodities—Aggregate		12%	6%	26%	85%	38%	84%	21%	14%	1%	4%
(P) Commodities—Gold					166%	154%	-18%	27%	13%	-1%	1%
(P) Commodities—Precious				28%	29%	185%	-27%	33%	11%	-2%	1%
(A) Trend—Equity	20%	23%	24%	77%	23%	-13%	13%	-3%	8%	11%	10%
(P) Commodities—Softs				-41%	243%	15%	11%	15%	8%	-3%	-1%
(A) Equity Factor—C-S Mom.	-15%	-18%	7%	35%	38%	44%	41%	26%	8%	4%	5%
(P) Commodities—Agri		12%	6%	-23%	197%	-21%	6%	33%	7%	-3%	0%
(A) Trend—FX					-14%	16%	42%	6%	4%	4%	4%
(A) Equity Factor—Quality (QMJ)				14%	-1%	-12%	40%	7%	3%	3%	3%
(P) Fixed Income—TIPS				-3%	13%	-2%	11%	6%	2%	3%	3%
(A) Equity Factor—Investment (CMA)				-7%	31%	-9%	24%	-10%	2%	2%	2%
(P) Long Equities—Energy Sector	-14%	-10%	25%	-19%	-19%	31%	31%	2%	1%	8%	6%

Neville, et. al. (2021, JPM)

미국 인플레이션 국면별 성과

□ 미국 인플레이션 국면별 성과

	Specific Inflation Regimes								Combined Regimes		
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)
Characteristics											
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007			
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008			
Length (months)	14	13	7	48	30	38	46	11			
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%			
Strategy	Real Return (total)								Real Return (annualized)		
(A) Equity Factor—Profitability (RMW)				4%	-24%	-8%	18%	6%	-1%	2%	2%
(A) Equity Factor—Value (HML)	-4%	-17%	3%	-8%	36%	-11%	-3%	-7%	-1%	2%	2%
(P) Real Estate—Residential	-17%	4%	-4%	-2%	-7%	11%	0%	-13%	-2%	2%	1%
(A) Equity Factor—Low Vol (BAB)	-24%	-6%	-3%	28%	-13%	9%	-7%	-22%	-3%	8%	6%
(P) Fixed Income—2-Yr. Treasury	-13%	-17%	-6%	-1%	-7%	-17%	11%	0%	-3%	2%	1%
(A) Equity Factor—Size (SMB)	-11%	-23%	-4%	45%	-43%	32%	-26%	-4%	-4%	1%	0%
(P) Fixed Income—10-Yr. Treasury	-11%	-17%	-6%	-13%	-12%	-31%	8%	5%	-5%	4%	2%
(P) Fixed Income—High Yield	-4%	-11%	0%	-18%	-21%	-38%	-10%	-8%	-7%	6%	4%
(P) Long Equities—Market Composite	-24%	-27%	24%	-7%	-48%	-14%	12%	-17%	-7%	10%	7%
(P) Fixed Income—Investment Grade	-7%	-12%	-3%	-23%	-20%	-43%	-5%	1%	-7%	6%	3%
(P) Fixed Income—30-Yr. Treasury	-17%	-17%	-6%	-20%	-28%	-41%	13%	2%	-8%	5%	3%
(P) Long Equities—Consumer Durables	-16%	-32%	24%	-30%	-62%	-27%	-28%	-36%	-15%	13%	7%

NOTES: The real total returns of assets analyzed in the article, through the eight US inflationary regimes shown in Exhibit 1 as well as the annualized return during inflationary, noninflationary, and all periods. In the first column, the strategy is denoted as active or passive by (A) or (P), respectively; returns for energies and gold in gray italics are spot returns for which we do not have futures data. These are not included in the combined regime calculation. The data vary by start date. Further details are provided in the body of this article and in Appendix A.

미국 CPI 바스켓 구성 요소

□ 미국 인플레이션 국면별 CPI 바스켓 구성 요소

EXHIBIT 3

Components of CPI Basket during US Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes	
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (82%)
Start Month	April 1941	March 1948	August 1950	February 1986	July 1972	February 1977	February 1987	September 2007		
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008		
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%		
Total Price Level Change (ann'd)	13%	19%	12%	4%	9%	10%	5%	6%		
Inflation Component	Average Price Rise in Regime (ann.)								Average Price Rise (ann.)	
Cereals and Bakery Products	9%	32%	16%	3%	20%	9%	6%	13%	13%	2%
Meat, Poultry, Fish, and Eggs					12%	10%	5%	4%	8%	2%
Dairy	15%	34%	23%	5%	12%	8%	5%	7%	14%	2%
Fruit and Veg	28%	10%	11%	4%	10%	9%	7%	12%	11%	3%
Soft Drinks			24%	2%	16%	14%	1%	3%	10%	3%
Other Food					22%	11%	3%	7%	11%	2%
Food Outside Home				6%	10%	10%	4%	5%	7%	3%
Alcohol				3%	6%	6%	4%	3%	5%	2%
Household Goods and Services					9%	7%	2%	1%	5%	1%
Apparel	20%	20%	17%	5%	6%	5%	5%	0%	10%	1%
New Vehicles				2%	5%	7%	2%	-1%	3%	2%
Medical Goods			4%	0%	3%	8%	8%	1%	4%	3%
Recreation Goods and Services								2%	2%	1%
Education Goods and Services								3%	3%	2%
Energy Products				3%	20%	25%	13%	53%	23%	1%
Energy Services	-1%	-1%	1%	1%	11%	11%	2%	18%	5%	2%
Tobacco Products			6%	6%	5%	6%	10%	8%	7%	6%
Shelter				6%	8%	13%	5%	3%	7%	3%
Water and Sewerage				4%	6%	8%	6%	5%	6%	5%
Medical Services	1%	11%	5%	8%	8%	10%	8%	4%	7%	4%
Transport Services	5%	4%	11%	6%	3%	8%	6%	6%	6%	3%
Personal Goods and Services			19%	4%	8%	7%	4%	3%	8%	2%

미국 주식 및 정부채의 실질 및 명목 수익률

□ 미국 인플레이션 국면별 미국 주식 및 정부채 성과

EXHIBIT 4

US Equity and Treasury Bond Performance across Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes				
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)	Hit Rate	t-Stat
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007					
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008					
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%					
Length (months)	14	13	7	48	30	38	46	11					
Strategy	Nominal Return (total)								Nominal Return (ann.)				
Equities	-13%	-12%	32%	10%	-32%	18%	35%	-12%	0%	12%	10%	50%	-3.2
Government Bonds	3%	0%	0%	4%	9%	-5%	30%	11%	3%	6%	5%	88%	-1.8
60-40	-7%	-7%	18%	8%	-17%	9%	35%	-3%	2%	10%	9%	50%	-3.6
Strategy	Real Return (total)								Real Return (ann.)				
Equities	-24%	-27%	24%	-7%	-46%	-14%	12%	-17%	-7%	10%	7%	26%	-4.8
Government Bonds	-11%	-17%	-6%	-13%	-12%	-31%	8%	5%	-5%	4%	2%	25%	-5.1
60-40	-19%	-23%	11%	-9%	-34%	-20%	12%	-8%	-6%	8%	6%	25%	-6.0

NOTES: Total returns to US equities, government bonds (10-year maturity), and a 60-40 equity-bond portfolio during the eight US inflationary regimes defined in the previous section, as well as the annualized return during inflationary, noninflationary, and all periods. In the final columns, we present the hit rate (proportion of inflationary periods with positive returns) and the t-statistic, which tests whether the returns in inflationary and noninflationary times are different. The data are from 1926 to 2020. See Appendix A for details on sources and methodology.

미국 섹터 성과

□ 미국 인플레이션 국면별 미국 섹터 성과

EXHIBIT 7

US Sector Performance in Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes				
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)	Hit Rate	t-Stat
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007					
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008					
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%					
Strategy	Real Return (total)								Real Return (ann.)				
Consumer Nondurables	-24%	-24%	11%	5%	-56%	-15%	42%	-7%	-6%	11%	8%	38%	-3.7
Consumer Durables	-16%	-32%	24%	-30%	-62%	-27%	-28%	-36%	-15%	13%	7%	13%	-5.0
Manufacturing	-23%	-23%	27%	-9%	-52%	-12%	-2%	-15%	-8%	11%	7%	13%	-3.7
Energy	-14%	-10%	25%	-19%	-19%	31%	31%	2%	1%	8%	6%	50%	-1.4
Chemicals	-25%	-17%	27%	-23%	-37%	-24%	9%	2%	-6%	11%	8%	38%	-3.5
Bus. Equip. (tech)	-26%	-34%	19%	37%	-58%	-11%	-29%	-15%	-9%	12%	8%	25%	-3.9
Telecoms	-32%	-25%	3%	-19%	-15%	-24%	41%	-27%	-7%	9%	6%	25%	-4.1
Utilities	-43%	-20%	12%	-24%	-38%	-22%	8%	-3%	-9%	10%	6%	25%	-4.5
Retail	-25%	-24%	19%	14%	-64%	-27%	6%	-15%	-9%	12%	8%	38%	-3.9
Health	-18%	-8%	18%	25%	-42%	-2%	43%	-6%	-1%	11%	9%	38%	-2.2
Financials	-20%	-29%	18%	23%	-53%	-5%	-24%	-35%	-9%	11%	7%	25%	-3.7
Other	-20%	-36%	25%	-12%	-62%	7%	-4%	-21%	-10%	8%	5%	25%	-3.1

NOTES: Total returns to 12 long-only sector portfolios, as collected on Kenneth R. French's website, during the eight US inflationary regimes defined in the previous section, as well as the annualized return during inflationary, noninflationary, and all periods. In the final columns, we present the hit rate (proportion of inflationary periods with positive returns) and the heteroskedasticity-consistent t-statistic, which tests whether the returns in inflationary and noninflationary times are different. The data are from 1926 to 2020. See Appendix A for details on sources and methodology.

채권 성과

□ 미국 인플레이션 국면별 채권

EXHIBIT 8

Fixed Income in Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes				
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)	Hit Rate	t-Stat
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007					
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008					
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%					
Strategy	Real Return (total)								Real Return (ann.)				
US Treasury 30 Yr.	-17%	-17%	-6%	-20%	-28%	-41%	13%	2%	-8%	5%	3%	25%	-5.0
US Treasury 10 Yr.	-11%	-17%	-6%	-13%	-12%	-31%	8%	5%	-5%	4%	2%	25%	-5.1
US Treasury 2 Yr.	-13%	-17%	-6%	-1%	-7%	-17%	11%	0%	-3%	2%	1%	13%	-5.8
USD Invest-Grd. Credit	-7%	-12%	-3%	-23%	-20%	-43%	-5%	1%	-7%	6%	3%	13%	-8.1
USD High-Yield Credit	-4%	-11%	0%	-18%	-21%	-38%	-10%	-8%	-7%	6%	4%	13%	-7.8
TIPS				-3%	13%	-2%	11%	6%	2%	3%	3%	60%	-0.6

NOTES: Total returns to 30-year, 10-year, and 2-year US Treasuries, investment-grade and high-yield credit, and 10-year TIPS during the eight US inflationary regimes defined in the previous section (see Exhibit 1), as well as the annualized return during inflationary, noninflationary, and all periods. In the final columns, we present the hit rate (proportion of inflationary periods with positive returns) and the heteroskedasticity-consistent t-statistic, which tests whether the returns in inflationary and noninflationary times are different. The data are from 1926 to 2020 for all assets other than TIPS, which are from 1959. See Appendix A for details on sources and methodology.

Neville, et. al. (2021, JPM)

상품 성과

□ 미국 인플레이션 국면별 상품

EXHIBIT 9

Commodities in Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes				
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)	Hit Rate	t-Stat
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007					
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008					
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%					
Strategy	Real Return (total)								Real Return (ann.)				
Industrials				115%	38%	-6%	306%	3%	19%	4%	7%	80%	1.7
Precious				28%	29%	185%	-27%	33%	11%	-2%	1%	80%	1.7
Agris		-12%	6%	-23%	197%	-21%	6%	33%	7%	-3%	0%	71%	1.8
Softs				-41%	243%	15%	11%	15%	8%	-3%	-1%	80%	1.6
Livestock				69%	-21%	35%	97%	-23%	7%	1%	2%	60%	1.1
Energies	-3%	2%	-6%	-16%	264%	57%	201%	68%	41%	-1%	3%	100%	1.7
Gold					166%	154%	-18%	27%	13%	-1%	1%	67%	1.6
Silver				9%	99%	210%	-41%	36%	12%	-5%	0%	80%	1.8
Commods. Aggregate		12%	6%	26%	85%	38%	84%	21%	14%	1%	4%	100%	3.1

NOTES: Total returns to six baskets of commodity futures, as well as gold and silver individually, and an equal-weighted, monthly rebalanced basket of all commodities during the eight US inflation regimes already defined, as well as the annualized return during inflationary, noninflationary, and all periods. Returns in gray italics are spot returns prior to the existence of a liquid futures contract. These are not included in the aggregate calculations. All other returns are funded. In the final columns, we present the hit rate (proportion of inflationary periods with positive returns) and the heteroskedasticity-consistent t-statistic, which tests whether the returns in inflationary and noninflationary times are different. The starting point for the data varies from 1946 (agris) to 1979 (energies) and runs through to 2020 at a monthly periodicity. See Appendix A for details on sources and methodology.

주거용 부동산과 수집품

□ 미국 인플레이션 국면별 주거용 부동산의 성과

EXHIBIT 11

Performance of Residential Real Estate in Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes				
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)	Hit Rate	t-Stat
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007					
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008					
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%					
Strategy	Real Return (total)								Real Return (ann.)				
Residential Real Estate	-17%	4%	-4%	-2%	-7%	11%	0%	-3%	-2%	2%	1%	25%	-5.1

EXHIBIT 12

Performance of Collectibles in Inflationary Regimes

Strategy	Real Return (total)								Real Return (ann.)			
Art	3%	-14%	-25%	98%	18%	54%	61%	-13%	7%	2%	3%	63%
Wine	156%	-39%	-6%	15%	-26%	120%	0%	-13%	5%	6%	6%	50%
Stamps	4%	-5%	-18%	35%	4%	204%	6%	16%	9%	3%	4%	75%

Neville, et. al. (2021, JPM)

주식 스타일과 추세 전략

□ 미국 인플레이션 국면별 미국 주식 스타일 성과

EXHIBIT 13

US Equity Style Performance in Inflationary Regimes

	Specific Inflation Regimes								Combined Regimes				
	US Enters WW2	End of WW2	Korean War	Ending of Bretton Woods	OPEC Oil Embargo	Iranian Revolution	Reagan's Boom	China Demand Boom	Inflation (19%)	Other (81%)	All (100%)	Hit Rate	t-Stat
Start Month	April 1941	March 1946	August 1950	February 1966	July 1972	February 1977	February 1987	September 2007					
End Month	May 1942	March 1947	February 1951	January 1970	December 1974	March 1980	November 1990	July 2008					
Total Price Level Change	15%	21%	7%	19%	24%	37%	20%	6%					
Strategy	Real Return (total)								Real Return (ann.)				
SMB—Size	-11%	-23%	-4%	45%	-43%	32%	-26%	-4%	-4%	1%	0%	25%	-1.8
HML—Value	-4%	-17%	3%	-8%	36%	-11%	-3%	-7%	-1%	2%	2%	25%	-1.5
RMW—Profitability				4%	-24%	-8%	18%	6%	-1%	2%	2%	60%	-1.5
CMA—Investment				-7%	31%	-9%	24%	-10%	2%	2%	2%	40%	-0.1
Momentum	-15%	-18%	7%	35%	38%	44%	41%	26%	8%	4%	5%	75%	0.6
QMJ—Quality-Junk				14%	-1%	-12%	40%	7%	3%	3%	3%	60%	-0.1
BAB—Betting against Beta	-24%	-6%	-3%	28%	-13%	9%	-7%	-22%	-3%	8%	6%	25%	-4.2
SMB—Size	-11%	-23%	-4%	45%	-43%	32%	-26%	-4%	-4%	1%	0%	25%	-1.8
HML—Value	-4%	-17%	3%	-8%	36%	-11%	-3%	-7%	-1%	2%	2%	25%	-1.5
RMW—Profitability				4%	-24%	-8%	18%	6%	-1%	2%	2%	60%	-1.5
CMA—Investment				-7%	31%	-9%	24%	-10%	2%	2%	2%	40%	-0.1
Momentum	-15%	-18%	7%	35%	38%	44%	41%	26%	8%	4%	5%	75%	0.6
QMJ—Quality-Junk				14%	-1%	-12%	40%	7%	3%	3%	3%	60%	-0.1
BAB—Betting against Beta	-24%	-6%	-3%	28%	-13%	9%	-7%	-22%	-3%	8%	6%	25%	-4.2